Catching a Moving Subspace: Low-Rank Bandits Beyond Stationarity

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Abstract

Many real bandit deployments — recommendation, clinical dosing, ad targeting — share two structural facts the linear-bandit literature has so far handled only in isolation: rewards live on a low-dimensional latent subspace, and that subspace drifts. Stationary low-rank bandits exploit the rank but break under any subspace change; non-stationary linear bandits adapt to drift but pay the ambient rate $\widetilde{\mathcal{O}}(d\sqrt{T})$ throughout. We study *piecewise-stationary low-rank* linear contextual bandits, in which the reward parameter factorizes as $\theta_t = B_k^\star w_t$ with a rank-r factor $B_k^\star \in \mathbb{R}^{d\times r}$ that is constant within each of K unknown segments and may shift at the boundaries, with only scalar reward feedback.

Our results are tight along three axes. **(i) Identification boundary.** With single-play scalar rewards, the moving subspace is recoverable through quadratic functionals of rewards if and only if three probe-side conditions hold (known noise variance, bounded state–noise coupling, full-dimensional probe support). Each is individually necessary in the unrestricted-second-moment problem, and the three are jointly sufficient — a characterization of the boundary of the solvable region, not just a sufficient interior point. **(ii) Algorithm and dynamic regret. SPSC** (Single-Play Subspace-Calibrated Optimism) interleaves isotropic probes with windowed projected ridge-UCB exploitation inside the learned r-dimensional subspace; a CUSUM-style adaptive variant discovers segment boundaries online. The costed dynamic regret is $\widetilde{\mathcal{O}}(r\sqrt{T}) + \widetilde{\mathcal{O}}(T^{2/3}) + O(W\,V_{\text{in}})$, replacing the ambient $d\sqrt{T}$ rate with the intrinsic rank. **(iii) Empirics.** On eleven benchmarks spanning synthetic, UCI/MovieLens, semi-synthetic clinical, and ZOZOTOWN production-log data, SPSC outperforms eleven non-stationary and low-rank baselines whenever $d-r \gtrsim T^{1/6}$, matching the analytical crossover; a direct necessity stress test confirms the identification boundary is not a proof artifact.

To our knowledge, this is the first work to characterize the identification boundary and to attain the intrinsic-rank dynamic-regret rate in the piecewise-stationary low-rank setting under scalar feedback.

1 Introduction

High-dimensional sequential decisions — personalized recommendation, clinical dosing, ad allocation — share two structural facts that the linear-bandit literature has so far treated only in isolation. The reward parameter $\theta_t \in \mathbb{R}^d$ at round t typically admits a *low-rank* factorization $\theta_t = B_k^\star w_t$ with $r \ll d$: user preferences, clinical responses, and ad-creative effects live on a thin slice of the ambient space. That thin slice is *not static*: tastes drift, cohorts change, treatment regimes get revised, and the subspace $\mathrm{range}(B_k^\star)$ itself moves across unknown change points $1 = \tau_0 < \tau_1 < \cdots < \tau_K = T{+}1$ that delimit the segments $\mathcal{I}_k = [\tau_{k-1}, \tau_k)$. Production deployments make both facts visible: Spotify's homepage contextual bandit explicitly targets daily shifts in user intent [Feijer et al., 2025]; the IWPC warfarin cohort [Consortium, 2009] has been re-analyzed to expose three latent patient subgroups whose dose–response coefficients flip sign across boundaries [Liu et al., 2025]; the public Open Bandit logs [Saito et al., 2021] ship scalar feedback collected across multiple ad campaigns at ZOZOTOWN. What is missing is a combined treatment in linear contextual bandits: rewards living on a low-rank subspace whose identity shifts at unknown change points, observed only through scalar feedback $y_t = x_t^\top \theta_t + \varepsilon_t$.

The gap, and why it is non-trivial. Stationary low-rank bandits [Jun et al., 2019, Jedra et al., 2024, Lu et al., 2021, Jang et al., 2024] deliver the $\widetilde{\mathcal{O}}(r\sqrt{T})$ rate but assume a fixed $B^\star$; any non-trivial subspace change voids the analysis ($\widetilde{\mathcal{O}}$ suppresses polylog factors in $T, d, r, 1/\delta$). Non-stationary linear bandits [Russac et al., 2019, Cheung et al., 2019] absorb drift through sliding windows or discounting but operate in $\mathbb{R}^d$ and pay $\widetilde{\mathcal{O}}(d\sqrt{T})$ within each segment, ignoring intrinsic rank entirely. Naive combinations fail in instructive ways: sliding-window ridge in $\mathbb{R}^d$ never recovers the subspace; a one-shot subspace estimate breaks the moment a change point fires; per-segment re-runs of stationary low-rank algorithms require oracle access to the boundaries. The genuine difficulty is twofold. (a) With scalar rewards, a single response y_t is a rank-one quadratic measurement of $\theta_t\theta_t^\top$, so the subspace is recoverable only through *second-order* probing. (b) Every probe diverts a round away from exploitation, so probe acquisition must itself be priced against the regret it enables. No prior algorithm recovers a changing subspace under scalar feedback, and no prior analysis prices that recovery against the exploitation regret it buys.

Contributions. We formulate the *piecewise-stationary low-rank* linear bandit (§2) and resolve it on three axes. **(i) Identification boundary (Theorem 2.2).** Three probe-side conditions (known noise variance, bounded state–noise coupling, full-dimensional probe support) are individually necessary in the unrestricted-second-moment problem and jointly sufficient for recovering $\mathrm{range}(B_k^\star)$ from quadratic functionals of scalar rewards. The necessity half — three matching impossibility results (Propositions B.12–B.14) — is what distinguishes this from a generic identifiability statement: it characterizes the boundary of the solvable region, not just a sufficient interior point. **(ii) Algorithm and costed dynamic regret (Theorem 4.1). SPSC** (Alg. 1) interleaves isotropic probes with windowed projected ridge-UCB exploitation inside the learned r-dimensional subspace via a quadratic-measurement identity; a **SPSC-Adaptive** variant (Alg. 2) uses a CUSUM-style detector to discover segment changes online. The costed dynamic regret is $\mathrm{DynReg}_T^{(c)} \leq \widetilde{\mathcal{O}}(r\sqrt{T}) + \widetilde{\mathcal{O}}(T^{2/3}) + O(W\,V_{\mathrm{in}})$, replacing the ambient $d\sqrt{T}$ rate with the intrinsic rank r, where W is the exploitation window and V_{in} the within-segment path variation. **(iii) Eleven benchmarks against eleven baselines.** A synthetic phase-transition grid, five UCI environments plus MovieLens, two semi-synthetic clinical datasets, the Russac small-d benchmark, and Open Bandit production logs, against LinUCB, D-LinUCB, SW-LinUCB, Restart-LinUCB, LowOFUL, VOFUL, LowRank-Reward, LinTS, SW-LinTS, and adapted stationary low-rank methods BOSS and Jedra. SPSC delivers regret reductions whenever $d - r \gtrsim T^{1/6}$, matching the analytical crossover. A direct necessity stress test (App. G.3, part C) restricts probe coverage to a proper subspace and reproduces the $\Omega(T)$ blow-up predicted by Proposition B.14, confirming the third identifiability condition is not a proof artifact.

To our knowledge, this is the first work to characterize the identification boundary and to attain the intrinsic-rank dynamic-regret rate in the piecewise-stationary low-rank setting under scalar feedback.

Related work. Stationary low-rank bandits [Jun et al., 2019, Kang et al., 2022, Jedra et al., 2024, Jang et al., 2024, Stojanovic et al., 2023, Duong et al., 2024, Lu et al., 2021] achieve intrinsic-rank rates but assume a fixed $B^\star$; nonstationary bandits [Russac et al., 2019, Cheung et al., 2019, Abbasi-Yadkori et al., 2023, Hou et al., 2024, Kim et al., 2022] bound dynamic regret in the ambient dimension and ignore low-rank structure; cost-aware observation [Seldin et al., 2014, Tucker et al., 2023, Elumar et al., 2024] prices information acquisition for fixed parameters, not for identifying a changing subspace. SPSC is the first algorithm we are aware of that (i) attains regret scaling with intrinsic rank r, (ii) adapts to changing subspaces without oracle boundaries, and (iii) prices probe acquisition explicitly. Appendix A gives the full positioning (Tables 5–7).

2 Setting and probe-based identification

Piecewise low-rank model. At every round t the reward parameter admits a factorization $\theta_t = B_k^\star w_t$, where $B_k^\star \in \mathbb{R}^{d\times r}$ has orthonormal columns and is piecewise constant over $K \geq 1$ segments $\mathcal{I}_k$ with unknown change points $1 = \tau_0 < \tau_1 < \cdots < \tau_K = T{+}1$, and the latent state $w_t \in \mathbb{R}^r$ follows a stable linear dynamical system inside each segment

$$w_t = A_k w_{t-1} + \eta_{t-1}, \qquad t \in \mathcal{I}_k,$$

with unknown A_k satisfying $\rho(A_k) \leq 1{-}\alpha_0$ for some $\alpha_0 > 0$ (used only to ensure $\|w_t\| \leq S_w$ via the stationary covariance bound) and zero-mean innovation η_{t-1} with covariance $\Sigma_{\eta,k} \succ 0$. We assume uniform action-boundedness $\sup_{x\in\mathcal{A}_t} \|x\| \leq R_{\mathcal{A}}$, a known probe distribution Q with $\mathbb{E}[uu^\top] \succeq \rho_Q I_d$ and sub-Gaussian marginals (bounded $\|u\| \leq L$ is a special case; see Remark 2.1), conditionally σ_ε-sub-Gaussian noise with unknown variance, bounded state-noise coupling $\|\mathbb{E}[\varepsilon_t\theta_t]\|_2 \leq \epsilon_\times$, and full-dimensional probe coverage. Theorem 2.2 below shows these three probe-side conditions are individually necessary and jointly sufficient for identifying the changing subspace from single-play scalar rewards. Throughout, $\delta \in (0,1)$ denotes the global confidence parameter (concentration bounds hold with probability $\geq 1{-}\delta$) and $\lambda > 0$ denotes the ridge regularizer used by the windowed estimator of §3.

Remark 2.1 (Probe distribution: scaled sphere)**.** *The theory uses scaled-sphere probes $u_t = \sqrt{d}\, v_t$, $v_t \sim \mathrm{Unif}(\mathbb{S}^{d-1})$, which satisfy $\|u_t\|_2 = \sqrt{d}$* exactly *(so no truncation event is needed) and $\mathbb{E}[uu^\top] = I_d$. In our implementation we draw $\tilde{u} \sim \mathcal{N}(0, I_d)$ and rescale $u \leftarrow \sqrt{d}\,\tilde{u}/\|\tilde{u}\|$, which is exactly the scaled-sphere distribution. The same argument applies to isotropic Gaussian probes $u_t \sim \mathcal{N}(0, I_d)$ via a χ^2_d truncation at $L = \sqrt{2d\log(2T/\delta)}$ (Appendix C).*

Quadratic measurement identity. Let $\widehat{\sigma}^2$ be the algorithm's estimate of the noise variance and define the centered probe statistic $s_t := y_t^2 {-} \widehat{\sigma}^2$ on every probe round $t \in \mathcal{T}_{\mathrm{probe}}$. Under the probe-noise assumptions of §2, for $\delta_\sigma := \widehat{\sigma}^2 - \sigma_\varepsilon^2$,

$$\mathbb{E}[s_t \mid \mathcal{H}_{t-1}, u_t] = u_t^\top \widetilde{M}_t u_t + 2u_t^\top \mathbb{E}[\varepsilon_t \theta_t \mid \mathcal{H}_{t-1}, u_t] - \delta_\sigma, \tag{1}$$

where $\widetilde{M}_t := \mathbb{E}[\theta_t\theta_t^\top \mid \mathcal{H}_{t-1}]$ is the predictable second moment, whose range is contained in $\mathrm{range}(B_k^\star)$ for $t \in \mathcal{I}_k$; under the non-degenerate innovation condition $\Sigma_{\eta,k} \succ 0$ (§2), the probe-time average has range exactly $\mathrm{range}(B_k^\star)$ (Proposition B.9). Under exact probe conditions ($\delta_\sigma = \epsilon_\times = 0$) this collapses to $\mathbb{E}[s_t \mid \cdot] = u_t^\top \widetilde{M}_t u_t$: *a scalar reward carries a quadratic measurement of $\widetilde{M}_t$.*

Lifted subspace estimator. The probe moment operator $\mathcal{K} : M \mapsto \mathbb{E}[(u^\top M u)\, uu^\top]$ admits a closed-form inverse on symmetric matrices for scaled-sphere probes $u = \sqrt{d}\, v$, $v \sim \mathrm{Unif}(\mathbb{S}^{d-1})$:

$$\mathcal{K}(M) = \tfrac{d}{d+2}\big(\mathrm{tr}(M)\, I_d + 2M\big), \qquad \mathcal{K}^{-1}(N) = \tfrac{d+2}{2d}\, N - \tfrac{\mathrm{tr}(N)}{2d}\, I_d, \qquad \|\mathcal{K}^{-1}\|_{\mathrm{op}\to\mathrm{op}} \leq 1 \tag{2}$$

on the symmetric subspace (Lemma B.4). Define the *lifted probe sample*

$$G_t := \mathcal{K}^{-1}(s_t\, u_t u_t^\top), \qquad \mathbb{E}[G_t \mid \mathcal{H}_{t-1}] = \widetilde{M}_t + \widetilde{B}_t, \tag{3}$$

with bias $\widetilde{B}_t = -(\delta_\sigma/d)\, I_d$ exactly on scaled-sphere probes, hence $\|\widetilde{B}_t\|_{\mathrm{op}} = |\delta_\sigma|/d$ (Lemma B.6). Averaging lifted samples across the probe rounds of segment k yields

$\widehat{M}_k := m_k^{-1} \sum_{t \in \mathcal{T}_k} G_t$ (with $\mathcal{T}_k := \mathcal{T}_{\text{probe}} \cap \mathcal{I}_k$ the probe rounds in segment k, $m_k := |\mathcal{T}_k|$), whose top-r eigenvectors form $\widehat{U}_k \in \mathbb{R}^{d \times r}$ with $\|\widehat{P}_k - P_k^\star\|_{\text{op}} \leq C_{\text{sub}} \sqrt{\log(d/\delta)/m_k}$ (Corollary B.10 below). The variance misspecification bias $\widetilde{B}$ is a scaled identity (population level), shifting all eigenvalues uniformly and *not rotating eigenvectors*: subspace recovery is robust to the plug-in estimate of σ_ε^2 at the population level. Finite-sample projector concentration still has a floor $\Delta_\sigma \propto |\widehat{\sigma}^2 - \sigma_\varepsilon^2|/(d\,\lambda_{\min})$ due to the random fluctuation around the biased mean (Cor. B.10).

Positive identification through the lifted moment G_t above and the necessity result that follows together yield the following theorem, the foundational result of the paper.

Theorem 2.2 (Identification under priced probing)**.** *Consider the model of §2 with scaled-sphere probes $u_t = \sqrt{d}\,v_t$, $v_t \sim \mathrm{Unif}(\mathbb{S}^{d-1})$, and the lifted probe sample $G_t = \mathcal{K}^{-1}(s_t u_t u_t^\top)$ from* (3)*.*

1. ***Identifiability.*** *Under (i) known noise variance σ_ε^2 ($\delta_\sigma = 0$), (ii) bounded state-noise coupling $\|\mathbb{E}[\varepsilon_t \theta_t]\|_2 \leq \epsilon_\times$, and (iii) full-dimensional probe support, the lifted sample satisfies $\mathbb{E}[G_t \mid \mathcal{H}_{t-1}] = \widetilde{M}_t$ exactly (the bias $\widetilde{B}_t = -(\delta_\sigma/d)\, I_d$ from Lemma B.6 vanishes under (i); the state-noise coupling term vanishes by sphere symmetry, regardless of $\epsilon_\times$), and the segment-averaged estimator $\widehat{M}_k = m_k^{-1} \sum_{t \in \mathcal{T}_k} G_t$ satisfies, with probability at least $1-\delta$, the projector concentration $\|\widehat{P}_k - P_k^\star\|_{\text{op}} \leq C_{\text{sub}} \sqrt{\log(d/\delta)/m_k}$; in particular $\widehat{U}_k = \mathrm{TopEig}_r(\widehat{M}_k)$ recovers $\mathrm{range}(B_k^\star)$ at rate $1/\sqrt{m_k}$ (Proof in Appendix B.2).*

2. ***Necessity (unrestricted-second-moment scope).*** *Each of (i), (ii), (iii) is individually necessary in the unrestricted second-moment problem: removing any one yields two distinct conditional second moments $\widetilde{M}_t \neq \widetilde{M}_t'$ that generate identical observation laws $\mathbb{E}[s_t \mid \mathcal{H}_{t-1}, u]$, so $\widetilde{M}_t$ is not identifiable from quadratic measurements. The three constructions are independent (each counterexample satisfies the remaining two conditions), so the three form a minimal identifiability set in this class. Within the rank-r LDS class of §2 ($r < d$), rank deficiency partially relaxes (i) since σ_ε^2 is identifiable from the $d - r$ smallest eigenvalues; (ii) and (iii) remain needed (Proof in Appendix B.3).*

3 Algorithm

SPSC (Alg. 1) alternates two phases:

- **Probe rounds** ($t \in \mathcal{T}_{\text{probe}}$): draw $u_t \sim Q$ (the probe distribution from §2; e.g. the scaled-sphere distribution of Remark 2.1), observe y_t, accumulate G_t in the current-segment estimator $\widehat{M}_t$, and refresh $\widehat{U}_t = \mathrm{TopEig}_r(\widehat{M}_t)$.
- **Exploitation rounds** ($t \notin \mathcal{T}_{\text{probe}}$): project each action $x \in \mathcal{A}_t$ to $z_t(x) = \widehat{U}_{t-1}^\top x \in \mathbb{R}^r$, maintain a *windowed* ridge-UCB estimate $\widehat{a}_t = \widetilde{V}_t^{-1} \widetilde{b}_t$ over a sliding window $\mathcal{W}_t$ of the last W exploitation rounds, and play $x_t = \arg\max_{x \in \mathcal{A}_t} \big\{ z_t(x)^\top \widehat{a}_t + \beta_t^{(r,W)} \|z_t(x)\|_{\widetilde{V}_t^{-1}} + \gamma_t \|x\|_2 \big\}$.

The confidence radius $\beta_t^{(r,W)} = \sigma_\varepsilon \sqrt{r \log(1 + W R_{\mathcal{A}}^2/(\lambda r)) + 2\log(2K/\delta)} + \sqrt{\lambda} S_w$ scales with $\sqrt{r}$, not $\sqrt{d}$: this is where the intrinsic-rank improvement enters. The correction term $\gamma_t = \Gamma_k + V_{k,t}(W)$ absorbs subspace-mismatch error $\Gamma_k \propto \varepsilon_k := \|\widehat{P}_k - P_k^\star\|_{\text{op}}$ and local within-window drift $V_{k,t}(W) := \sum_{s, s+1 \in \mathcal{I}_k,\, t-W \leq s < t} \|\theta_{s+1} - \theta_s\|_2$ (formal definitions in App. B.4). Per-round cost is $O(dr|\mathcal{A}_t| + r^2)$ vs. $O(d^2|\mathcal{A}_t|)$ for ambient LinUCB, an $O(d/r)$ speed-up in addition to the statistical gain.

Full pseudocode for Alg. 1 (oracle boundaries) is in Appendix D, alongside the adaptive variant described next.

Adaptive variant (unknown boundaries). SPSC-Adaptive removes the oracle-boundary assumption by detecting segment changes online: it maintains two non-overlapping rolling-window estimators $\widehat{M}_t^{\text{recent}}, \widehat{M}_t^{\text{past}}$ of the probe-time second moment and triggers a subspace reset whenever $S_t := \|\widehat{M}_t^{\text{recent}} - \widehat{M}_t^{\text{past}}\|_{\text{op}}$ exceeds a calibrated threshold b. The threshold is chosen so that, with probability $1-\delta_{\text{FA}}$, no false alarm occurs across the horizon, while every true change of size $\Delta_k \geq 2b$ is detected within a bounded delay $D_{\max}$ (specified in Appendix D together with the full pseudocode, tuning, and CUSUM analysis).

4 Regret guarantees

Balanced probe schedule. The probe schedule is *balanced*: there is an absolute constant a_0 such that, for every segment $k \in \{1, \ldots, K\}$ and every prefix-count $j \in \{0, 1, \ldots, m_k\}$, the number of exploit rounds in E_k with exactly j probes collected before them is at most $a_0\lceil \ell_k/m_k \rceil$. (Uniformly spaced probes satisfy this.)

Theorem 4.1 (Costed dynamic regret of Alg. 1)**.** *Run Alg. 1 with oracle segment boundaries $\{\mathcal{I}_k\}_{k=1}^K$, scaled-sphere probes, and a balanced probe schedule. Under the model of §2, Assumption B.2, and exact centering $\widehat{\sigma}^2 = \sigma_\varepsilon^2$, set the per-segment probe budget to $m_k = \min\{\ell_k, \lceil c_0 \ell_k^{2/3} \rceil\}$ for an absolute constant $c_0 > 0$ (chosen as the closed-form optimum* (13)*; segments shorter than the burn-in threshold $q^\star = O(\log(KdT/\delta)/\lambda_{\min}^2)$ are charged trivially), where $\lambda_{\min} > 0$ is the predictable probe-time rth-eigenvalue lower bound (Lem. B.1). Then with probability at least $1-\delta$ and for fixed K,*

$$\mathrm{DynReg}_T^{(c)} \;\leq\; \widetilde{\mathcal{O}}\big(r\sqrt{T}\big) \;+\; \widetilde{\mathcal{O}}\big(T^{2/3}\big) \;+\; O\big(WV_{\text{in}}\big). \tag{4}$$

Including all constants: $\mathrm{DynReg}_T^{(c)} \leq \widetilde{\mathcal{O}}(r\sqrt{KT}) + \widetilde{\mathcal{O}}(A^{1/3}B^{2/3}K^{1/3}T^{2/3}) + O(R_{\mathcal{A}}WV_{\text{in}}) + \widetilde{\mathcal{O}}(K^{2/3}T^{1/3}/\lambda_{\min}^2)$*, with* $A := (R_{\mathcal{A}} + R_Q)S_w + c$*,* $B := CR_{\mathcal{A}}S_w(1 + R_{\mathcal{A}}\sqrt{W/\lambda})\sqrt{\log(2KdT/\delta)}$*,* $R_Q := \sup_{u\in\text{supp}(Q)}\|u\|_2$ *($R_Q = \sqrt{d}$ for scaled-sphere; $R_Q \leq R_{\mathcal{A}}$ if probes lie inside the action set); the $\widetilde{\mathcal{O}}(K^{2/3}T^{1/3}/\lambda_{\min}^2)$ summand is the burn-in cost (lower order than $\widetilde{\mathcal{O}}(T^{2/3})$ for fixed K). Here $V_{\text{in}} := \sum_k \sum_{s,s+1\in\mathcal{I}_k} \|\theta_{s+1} - \theta_s\|_2$ is the within-segment path variation and W is the exploitation window (cumulative on E_k when $W \geq \ell_k$); the bound is most informative when V_{in} is small. Proof in §B.5.*

Corollary 4.2 (Plug-in variance)**.** *Estimate $\widehat{\sigma}^2$ from N probe-round residuals (Lem. B.11) and write $\delta_\sigma := |\widehat{\sigma}^2 - \sigma_\varepsilon^2|$. For scaled-sphere probes, the bias $\widetilde{B} = -(\delta_\sigma/d)I_d$ from Lem. B.6 is a scaled identity, hence preserves the eigenvectors of $\bar{M}_k^{\text{probe}}$. Davis–Kahan applied with the biased mean as reference removes the Δ_σ floor at population level, so Theorem 4.1 holds with the displayed bound, $\lambda_{\min}$ replaced by $\lambda_{\min} - \delta_\sigma/d$ in the constants. Choosing $N \gtrsim T^{2/3}/d^2$ yields $\delta_\sigma/d \ll \lambda_{\min}$ with high probability, so the leading rate is preserved.*

Corollary 4.3 (Within-segment-stationary special case)**.** *Suppose $V_{\text{in}} = 0$, i.e. within each segment $\mathcal{I}_k$ the parameter is constant ($\theta_t \equiv \theta^{(k)}$ for $t \in \mathcal{I}_k$, with $\theta^{(k)} \in \mathrm{range}(B_k^\star)$). The non-degenerate innovation assumption ($\Sigma_{\eta,k} \succ 0$) still holds* across *the segment family, so the rank-r subspace* $\mathrm{range}(B_k^\star)$ *is well-defined and identifiable through SPSC's probe channel. The bound of Theorem 4.1 reduces to*

$$\mathrm{DynReg}_T^{(c)} \;\leq\; \widetilde{\mathcal{O}}(r\sqrt{T}) + \widetilde{\mathcal{O}}(T^{2/3})$$

for fixed K (suppressing polylog factors). The leading $\widetilde{\mathcal{O}}(r\sqrt{T})$ term comes from LinUCB on the r-dimensional projected target $z_t^\top a^{(k)}$ (this does not require the within-segment target to itself span an r-dimensional set; the projector is what collapses the geometry to dimension r, not the trajectory of θ_t). This is the regime that class- or cohort-induced segment shifts approximate empirically.

Scope of the bound. Theorem 4.1 analyzes exactly the practical Alg. 1: probes are interleaved according to a balanced schedule, $\widehat{U}_t$ is refreshed online, and exploitation rounds re-project the windowed history through the current basis. The time-varying-basis subspace error $\varepsilon_{k,t} = \|\widehat{P}_t - P_k^\star\|_{\text{op}}$ is controlled uniformly in t via a prefix concentration argument (Lemma B.17).

The leading $\widetilde{\mathcal{O}}(r\sqrt{T})$ term beats the ambient baseline $\widetilde{\mathcal{O}}(d\sqrt{T})$ whenever $r \ll d$; the $\widetilde{\mathcal{O}}(T^{2/3})$ term is the price of acquiring the subspace information online. The constant B depends on the exploitation window W as $\sqrt{W}$; for W treated as a problem parameter (the setting throughout this paper, consistent with $W \leq \min_k \ell_k$ on ℓ_k-bounded segments), this is absorbed into the constants and the $\widetilde{\mathcal{O}}(T^{2/3})$ rate is preserved. If W is allowed to scale with T, the second term becomes $\widetilde{\mathcal{O}}(W^{1/3}T^{2/3})$. Solving $r\sqrt{T} \lesssim T^{2/3}$ gives a simple rule of thumb: SPSC tends to outperform ambient LinUCB whenever $d - r \gtrsim T^{1/6}$. The experiments in §5 show a crossover qualitatively consistent with this rule of thumb.

Corollary 4.4 (Rank-adaptive guarantee)**.** *If the true rank r is unknown, thresholding $\widehat{M}_k$ at $\tau_k^{\mathrm{rank}} := 2R_X\sqrt{\log(2d/\delta)/m_k}$ (R_X the lifted-sample envelope from App. B.2) recovers the true rank with probability $\geq 1-\delta$ whenever the population eigengap exceeds $4\tau_k^{\mathrm{rank}}$, and the bound* (4) *applies unchanged. Proof in §B.6.*

Proposition 4.5 (Adaptive SPSC)**.** *Suppose every true change satisfies $\Delta_k \geq 2b$ for the threshold b set in Appendix B.7. Then with probability at least $1 - \delta_{\mathrm{FA}}$, SPSC-Adaptive (Alg. 2) attains the same leading rate as Theorem 4.1 with an additive $O(KW_{\mathrm{det}})$ delay overhead. Proof in Appendix B.7.*

5 Experiments

We evaluate SPSC and SPSC-Adaptive on eleven benchmarks: a synthetic phase-transition grid; UCI/MovieLens (Covertype, Pendigits, Satimage, MNIST, Fashion-MNIST, MovieLens); clinical (Warfarin, Vancomycin); the Russac small-d stress test (§5.4); and Open Bandit production logs (§5.5). Baselines: LinUCB [Abbasi-Yadkori et al., 2011], D-LinUCB [Russac et al., 2019], SW-LinUCB [Cheung et al., 2019], Restart-LinUCB, LowRank-Reward, LowOFUL [Jun et al., 2019], VOFUL [Kim et al., 2022], LinTS, SW-LinTS, and adapted BOSS [Duong et al., 2024]/Jedra [Jedra et al., 2024]; Oracle-LinUCB (true subspace) is an optimistic reference. Curves are means over ≥ 10 seeds with ± 1 SE bands.

5.1 Phase-transition boundary

Theory predicts a crossover at $d - r \asymp T^{1/6}$: SPSC wins above it, ambient LinUCB below. We verify on a piecewise low-rank LDS at T=5,000, sweeping $(d, r) \in \{5, 10, 20, 30, 45, 60, 80, 100\} \times \{1, 3, 5, 10, 15, 20\}$ with $r<d$ (40 cells), K=10, σ_ε=0.3, spectral radius 0.99, 40 actions, probe cost c=0.1.

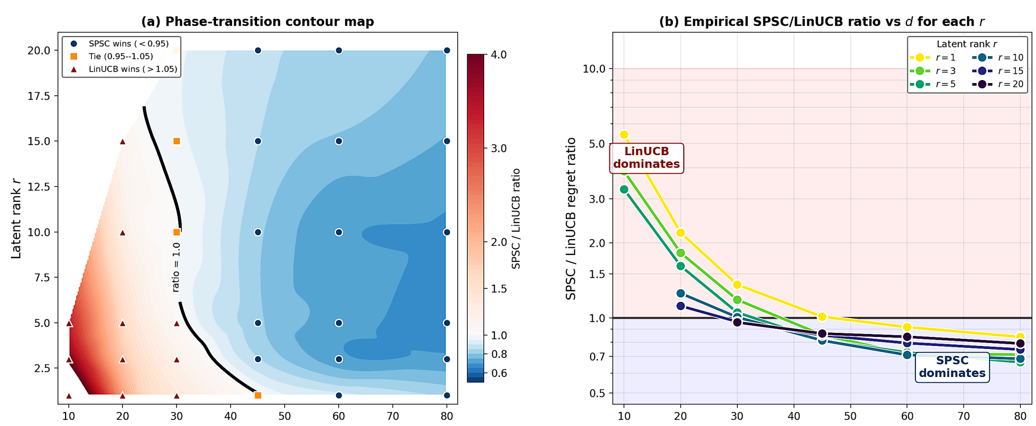

Figure 1: **Phase-transition boundary.** SPSC/LinUCB costed-regret ratio across the (d, r) grid. The crossover $d - r = T^{1/6}$ (dashed) separates SPSC-wins (ratio <1) from ambient-wins; per-rank curves track $\sqrt{r/d}$ (dotted), matching Theorem 4.1.

SPSC achieves 13–34% reductions whenever $d \geq 45$ with every r-curve tracking $\sqrt{r/d}$: use SPSC when $d - r \gtrsim T^{1/6}$, and ambient LinUCB otherwise.

5.2 Real-data multi-baseline comparison

Six real-data benchmarks recast as piecewise low-rank bandits on a (d, r)-grid: UCI **Covertype** (K=4, T=10,000); **Pendigits**, **Satimage**, pixelized **MNIST** and **Fashion-MNIST** (random-projection features), and **MovieLens-100K** with genuine user ratings (all K=10, T=5,000, 10 seeds). Segments are class or user-subset shifts. The first five have cluster-induced low-rank structure; MovieLens is only approximately low-rank (rank-d SVD features, §E) and serves as a robustness stress test. Table 1 reports SPSC vs. LinUCB and vs. best non-oracle competitor on representative cells; per-cell tables in Appendix F.

Table 1: **SPSC/LinUCB** and **SPSC-Adaptive/Best Competitor** ratios across six real-data benchmarks (10 seeds per cell). Values <1 indicate SPSC wins; **bold** marks cells where an SPSC variant beats every non-oracle baseline. SPSC trails LinUCB at $d \approx r$ (top of each block) and gains as d grows relative to r. Restart-LinUCB, the closest competitor, is subspace-oblivious, so its regret is constant along each row.

	d=55		d=105		d=155/200		best	
Dataset / r	S/Lin	S/Best	S/Lin	S/Best	S/Lin	S/Best	cell	ratio
Covertype, r=10	0.96	0.99	**0.88**	**0.92**	—	—	(105, 20)	**0.86**
Pendigits, r=10	**0.76**	**0.77**	**0.65**	**0.73**	—	—	(105, 10)	**0.65**
Satimage, r=10	**0.47**	**0.59**	**0.40**	**0.58**	—	—	(105, 10)	**0.40**
MNIST, r=10	**0.84**	**0.86**	**0.74**	**0.88**	**0.73/0.98**	**0.98**	(105, 20)	**0.74**
Fashion-MNIST, r=10	**0.88**	**0.90**	**0.73**	**0.88**	**0.67**/0.99	0.99	(105, 20)	**0.71**
MovieLens, r=10	1.00	1.00	**0.97**	**0.97**	**0.92/0.93**	**0.93**	(200, 5)	**0.92**

At (d=105, r=10), SPSC-Adaptive beats LinUCB by −47% on Pendigits (operating-regime grid in Fig. 2, App. F) and −68% on Satimage (closest competitor VOFUL is +112% above SPSC-Adaptive; per-cell results in Table 10, App. F); at (d=105, r=20) on Covertype it beats Restart-LinUCB by −15%. Non-low-rank baselines are by construction r-insensitive.

5.3 Clinical benchmarks: Warfarin and Vancomycin dosing

Warfarin. Warfarin dosing depends on demographics, genotype (VKORC1, CYP2C9), and clinical indicators [Consortium, 2009]. We calibrate to the IWPC cohort (∼5,700 patients, d=93, K=8 segments, T=5,000, 10 seeds; reward = negative deviation from therapeutic dose). Table 2 (r=3): SPSC-Adaptive cuts regret by 67.3% vs. LinUCB; full per-rank tables in Table 14 (App. F); random-subspace ablation in App. I.

Vancomycin. With the AUC-targeted Rybak et al. [2020] protocol (Cockcroft–Gault CrCl $\to$ $\mathrm{CL}_{\mathrm{vanco}} = 0.04\,\mathrm{CrCl}$, daily dose at AUC_{24}=500; d=93, K=8, T=5,000, 10 seeds), Table 3 sweeps $r \in \{1, 2, 3, 5, 10\}$: SPSC-Adaptive cuts regret by 36.4% at r=1 and 14.5–25.7% for $r \in \{2, 3, 5, 10\}$ vs. LinUCB, beating every non-oracle baseline at every rank. Per-rank tables in Appendix K.

5.4 Standard nonstationary benchmark

On the small-d stress test of Russac et al. [2019] (d=2, r=1, K=4, T=6,000, 50 arms, 30 seeds), SPSC reduces regret by −**46.5**% vs. D-LinUCB, −**34.8**% vs. SW-LinUCB, and −**52.0**% vs. stationary OFUL (itself +11.5% above D-LinUCB here). Full table in Appendix H.

5.5 Real exploration logs: Open Bandit Pipeline (ZOZO)

The Open Bandit Dataset [Saito et al., 2021] ships production exploration logs (random-policy ZOZOTOWN slice, T=5,000, K=10, 10 seeds); only approximately low-rank, so this is a *robustness* stress test rather than a setting where the rank-r assumption holds exactly. At d=55, r=5, SPSC reaches $\mathbf{83 \pm 2}$ vs. LinUCB's 102 ± 1 (−18.8%); SPSC-Adaptive −10.9%. SPSC neither breaks nor over-claims on weakly-low-rank production data. Per-cell results in Table 15 (App. F).

Table 2: **Warfarin** (representative cell r=3, mean±SE, 10 seeds). SPSC-Adaptive cuts regret by 67.3% vs. LinUCB. Full per-rank tables in Appendix F.

Method	Costed regret	vs. LinUCB
Oracle LinUCB	177 ± 4	−91.2%
SPSC Alg. 1 (ours)	1372 ± 33	−31.6%
SPSC-Adaptive (ours)	$\mathbf{657 \pm 39}$	**−67.3%**
LowOFUL	683 ± 97	−66.0%
VOFUL	779 ± 93	−61.2%
LowRank-Reward	1056 ± 40	−47.4%
SW-LinUCB	2096 ± 18	+4.5%
LinUCB	2006 ± 17	—

Table 3: **Vancomycin** (mean±SE, 10 seeds): SPSC-Adaptive control regret across range of r vs. LinUCB and the best non-oracle baseline. Full tables in Appendix K.

r	SPSC-Adaptive	vs. LinUCB	vs. best comp.
1	$\mathbf{525.3 \pm 45.8}$	−36.4%	−35.1%
2	$\mathbf{613.6 \pm 33.1}$	−25.7%	−17.1%
3	$\mathbf{640.3 \pm 33.0}$	−22.4%	−10.1%
5	$\mathbf{665.9 \pm 27.2}$	−19.3%	−12.7%
10	$\mathbf{705.6 \pm 19.5}$	−14.5%	− 7.0%

5.6 Comparison with stationary low-rank methods (oracle restarts)

For a best-effort comparison with stationary low-rank methods, we adapt BOSS [Duong et al., 2024] and Jedra [Jedra et al., 2024] by restarting at every true segment boundary, an oracle advantage we do not give SPSC (setup in Appendix J). Table 4: SPSC wins all eight cells by 5.8–19.9%, with the lead largest at small d (−16.3 to −19.9% at d=55) and narrowing to 5.8–8.0% at d=200. SPSC-Adaptive (no oracle) also beats both on every cell.

Table 4: **Adapted stationary low-rank methods on a (d, r)-grid**. Costed regret (mean±SE, 10 seeds, T=5,000, K=10). BOSS/Jedra restart at *oracle* segment boundaries; SPSC-Adaptive detects them online. SPSC variants win every cell.

d	r	SPSC Alg. 1	SPSC-Adaptive	BOSS (oracle)	Jedra (oracle)	SPSC vs. best comp.
55	5	$\mathbf{323 \pm 10}$	342 ± 15	399 ± 17	386 ± 18	−16.3%
55	10	$\mathbf{471 \pm 10}$	507 ± 13	588 ± 27	602 ± 24	−19.9%
105	5	$\mathbf{264 \pm 9}$	274 ± 10	300 ± 14	299 ± 13	−11.7%
105	10	$\mathbf{377 \pm 15}$	401 ± 14	423 ± 18	429 ± 14	−10.9%
105	20	$\mathbf{578 \pm 12}$	608 ± 14	654 ± 17	654 ± 14	−11.6%
200	5	$\mathbf{208 \pm 5}$	218 ± 6	236 ± 9	226 ± 9	− 8.0%
200	10	$\mathbf{277 \pm 10}$	292 ± 12	294 ± 11	305 ± 15	− 5.8%
200	20	$\mathbf{442 \pm 13}$	452 ± 11	473 ± 16	489 ± 18	− 6.6%

Robustness of SPSC-Adaptive. Detailed comparison under correct vs. misspecified oracle boundaries is in Appendix L: SPSC-Adaptive is robust to false-alarm boundaries, while Alg. 1 degrades monotonically as the oracle is corrupted.

5.7 Rank misspecification, probe-rate sensitivity, and take-away

Rank misspecification (Covertype, d=155, $r^\star$=10; App. G.2): sweeping over nine grid points $r \in \{1, 3, 5, 10, 15, 20, 30, 50, 80\}$ gives a U-shaped curve (under: +33% at r=1; sweet spot r=30, −10%); SPSC beats LinUCB across $r \in [15, 50]$, so mild overestimation is safe. **Probe rate** (App. G.1)**:** on the synthetic reference setting (d=4, r=1), the probe-period sweep $\{5, 10, 20, 30, 50, 100, 300\}$ has its optimum near 20–30, matching $m_k^\star \propto \ell_k^{2/3}$. **Additional datasets** (see Appendix F): SPSC wins every cell at $d \geq 105$, peaking at (d, r)=(105, 20). **Sensitivity & dynamics.** Large-scale assumption sweeps (App. G.4) confirm SPSC tracks Oracle within a constant factor through variance misspecification, approximate-rank perturbation ($\epsilon_k^{\perp}$ up to 0.4), and spectral radius up to ρ=0.95; SPSC remains the best non-Oracle method at ρ=0.99. Subspace error tracks the predicted $1/\sqrt{m}$ rate (App. G.5); noise/change-point-frequency/drift-speed sweeps (App. G.6) confirm the same trends. **Take-away.** Across the eleven benchmarks, SPSC delivers $\sqrt{r/d}$ intrinsic-rank savings whenever $d - r \gtrsim T^{1/6}$ and does not catastrophically fail outside that regime.

6 Conclusion

Sequential decision problems in recommendation, clinical dosing, and adaptive experimentation share two features: rewards live on a low-rank subspace of rank $r \ll d$, and that subspace shifts at unknown change points. Stationary low-rank, nonstationary, and cost-aware bandits each cover one piece; none recover a changing subspace under priced single-play scalar feedback. We close this gap: three probe-side conditions, known noise variance, bounded state-noise coupling, and full-dimensional probe support, are each individually necessary (in the unrestricted-second-moment scope) and jointly sufficient for subspace recovery from quadratic functionals of scalar rewards (Theorem 2.2). Based on this, we propose SPSC, which interleaves probing with windowed projected ridge-UCB in the learned r-dimensional subspace and achieves costed dynamic regret $\widetilde{\mathcal{O}}(r\sqrt{T}) + \widetilde{\mathcal{O}}(T^{2/3}) + O(WV_{\text{in}})$, replacing the ambient $d\sqrt{T}$ rate with the intrinsic r (Theorem 4.1); a CUSUM adaptive variant attains the same rate without oracle change points (Proposition 4.5). On eleven benchmarks across synthetic, UCI, clinical, and production data, SPSC reduces regret whenever $d - r \gtrsim T^{1/6}$, a crossover consistent with theory.

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Appendix: Table of Contents

A Comparison with prior work

This appendix positions SPSC against published work in low-rank bandits, nonstationary bandits, and active-query bandits via three tables: Table 5 (head-to-head with seven recent works on regret form, change-point adaptation, and probing budget), Table 6 (compact positioning of SPSC's combined setting against the three closest *problem classes*), and Table 7 (regret-rate comparison with five published baselines that appear in our experiments). The take-away is that the existing literature splits along three axes (low-rank, nonstationary, costly probing) and no prior method addresses all three; SPSC is the first to do so with a self-contained identification chain that turns rank-one scalar rewards into matrix-valued evidence.

B Proofs

This section proves the statements of §4 in dependency order:

1. §B.1 fixes notation and establishes the probe-time excitation lemma that all proofs depend on.
2. §B.2 derives the load-bearing *identification chain*: closed-form inverse of the probe operator $\mathcal{K}$, quadratic-measurement identity, lifted probe sample's near-unbiasedness, matrix Bernstein/Freedman, and the projector error via Davis–Kahan.
3. §B.4 states and proves the interleaved-analysis lemmas: projected nonstationary UCB with time-varying basis (Lem. B.16), uniform prefix subspace concentration (Lem. B.17), and cumulative interleaved subspace error (Lem. B.18).

Table 5: Comparison with related literature. "Prob." indicates a probabilistic regret bound; "CP?" indicates change-point adaptation.

Reference	**Venue/ Year**	**Regret / Theory**	**Prob.**	**CP?**	**Difference vs. ours**
Kang et al. [2022]	NeurIPS'22	$\widetilde{\mathcal{O}}\left(\sqrt{(d_1+d_2)MrT}\right)$	✓	✗	Stationary; probing not budgeted.
Stojanovic et al. [2023]	NeurIPS'23	Entry-wise / subspace recovery	✗/✓	✗	Estimation focus; no probe budget.
Jedra et al. [2024]	ICML'24	$\widetilde{\mathcal{O}}\left(r^{5/4}(m+n)^{3/4}\sqrt{T}\right)$	✓	✗	Stationary; no re-learning.
Jang et al. [2024]	ICML'24	Arm-set-dependent regret	✓	✗	Stationary; no CP detection.
Abbasi-Yadkori et al. [2023]	JMLR'23	$\widetilde{\mathcal{O}}(\sqrt{KN(S+1)})$	✗	✓	Unstructured K-arm; no low-rank.
Komiyama et al. [2024]	JMLR'24	ADR-bandit finite-time	✗	✓	Unstructured; no low-rank + probing.
Hou et al. [2024]	NeurIPS'24	Fixed-confidence sample complexity	✓	✓	BAI objective; no low-rank.

Table 6: Compact positioning of SPSC's combined setting.

Work class	**Low-rank**	**Nonstationary**	**Costly probing**	**Main gap vs. ours**
Low-rank bandits	✓	✗	✗	Fixed subspace; no piecewise changes; no sensing-cost budget.
Nonstationary linear/contextual bandits	✗	✓	✗	Adapt in ambient parameter space; ignore latent low-rank structure.
Costly-information / active-query bandits	✗	✗	✓	Price information acquisition; do not address subspace recovery under changing representations.
This paper	✓	✓	✓	**Combines low-rank structure, piecewise-stationary adaptation, and explicit costly probing in a single identification-to-exploitation framework.**

Table 7: Regret-scaling comparison with the published baselines that appear in our experiments.

Algorithm	Reference	Regret scaling	Key difference vs. SPSC
LinUCB	Abbasi-Yadkori et al. [2011]	$O(d\sqrt{T})$	Ambient; no subspace exploitation.
LowOFUL	Jun et al. [2019]	$\widetilde{\mathcal{O}}(r\sqrt{T})$	Needs oracle subspace B.
LowRank-Reward	Lu et al. [2021]	$\widetilde{\mathcal{O}}(r\sqrt{T})$	Stationary; learns B from rewards.
BOSS	Duong et al. [2024]	$\widetilde{\mathcal{O}}(r\sqrt{dT})$	Stationary; no online change adaptation.
Jedra	Jedra et al. [2024]	$\widetilde{\mathcal{O}}(r^{5/4}(m+n)^{3/4}\sqrt{T})$	Stationary; no probe cost.
SPSC (ours)	This work	$\widetilde{\mathcal{O}}(r\sqrt{T}+T^{2/3})$	Learns a *changing* B via priced probes.

4. §B.5 proves the main costed regret bound (Thm. 4.1) via a five-step decomposition: probe regret, exploitation regret via Lem. B.16, per-segment bound, probe-estimation tradeoff with burn-in, and sum over segments.
5. §B.6 derives the rank-adaptive corollary (Cor. 4.4) via Weyl's inequality on the matrix-Bernstein bound.
6. §B.7 lifts the guarantee to SPSC-Adaptive (Prop. 4.5) by controlling the CUSUM detector.

The windowed projected self-normalized bound (Lem. B.3), the windowed projected potential condition (Assumption B.2), and the projected nonstationary UCB lemma (Lem. B.16) are the linear-bandit ingredients; constants that do not affect leading order are absorbed into $\widetilde{\mathcal{O}}(\cdot)$.

B.1 Preliminaries and notation

Throughout, $\mathcal{I}_k = [\tau_{k-1}, \tau_k)$ denotes segment k of length $\ell_k := |\mathcal{I}_k|$, so $\sum_k \ell_k = T$. Inside $\mathcal{I}_k$ the learner performs $m_k = |\mathcal{T}_k|$ probe rounds ($\mathcal{T}_k := \mathcal{T}_{\text{probe}} \cap \mathcal{I}_k$) and $n_k = \ell_k - m_k$ exploitation rounds ($E_k := \mathcal{I}_k \setminus \mathcal{T}_k$). We write $\widehat{P}_t = \widehat{U}_t \widehat{U}_t^\top$ and $P_k^\star = B_k^\star B_k^{\star\top}$ for the estimated and true rank-r projectors, and $\varepsilon_k := \|\widehat{P}_k - P_k^\star\|_{\text{op}}$. We work on the high-probability event $\mathcal{G}_w := \{\max_{t \le T} \|w_t\|_2 \le S_w\}$. For sub-Gaussian innovations one may take $S_w = \widetilde{\mathcal{O}}(\sqrt{r})$ via union bound (the failure probability is absorbed into the overall δ budget); for bounded innovations $\mathcal{G}_w$ holds deterministically. In what follows we implicitly condition on $\mathcal{G}_w$.

The *predictable second moment* is $\widetilde{M}_t := \mathbb{E}[\theta_t \theta_t^\top \mid \mathcal{H}_{t-1}] = B_k^\star \mathbb{E}[w_t w_t^\top \mid \mathcal{H}_{t-1}](B_k^\star)^\top$, and the probe-time average is $\bar{M}_k^{\text{probe}} := m_k^{-1} \sum_{t \in \mathcal{T}_k} \widetilde{M}_t$. By the non-degenerate-innovation assumption $\Sigma_{\eta,k} \succ 0$ of §2, $\bar{M}_k^{\text{probe}}$ has rank r and its r-th eigenvalue $\lambda_{\min} > 0$ is uniformly bounded below in k (Lem. B.1 below; stability of A_k is used only to bound $\|w_t\| \le S_w$, not for this lower bound). Constants $C, C', \dots$ depend only on $\sigma_\varepsilon, R_{\mathcal{A}}, L, \rho_Q, S_w, \lambda_{\min}$ and may change line to line.

Lemma B.1 (Predictable probe-time excitation)**.** *For each segment k,*

$$\lambda_r\left(\frac{1}{m_k} \sum_{t \in \mathcal{T}_k} \mathbb{E}[w_t w_t^\top \mid \mathcal{H}_{t-1}]\right) \;\ge\; \lambda_{\min} \;>\; 0 \qquad \textit{almost surely.}$$

Proof. For each $t \in \mathcal{I}_k$, the conditional second moment satisfies $\mathbb{E}[w_t w_t^\top \mid \mathcal{H}_{t-1}] = A_k w_{t-1} w_{t-1}^\top A_k^\top + \Sigma_{\eta,k} \succeq \Sigma_{\eta,k}$ (LDS recursion + non-degenerate innovations). Hence $\lambda_r(\mathbb{E}[w_t w_t^\top \mid \mathcal{H}_{t-1}]) \ge \lambda_r(\Sigma_{\eta,k}) =: \lambda_{\min} > 0$ uniformly across rounds. Averaging over any probe subset $\mathcal{T}_k \subseteq \mathcal{I}_k$ preserves this PSD lower bound: $\lambda_r(m_k^{-1} \sum_{t \in \mathcal{T}_k} \mathbb{E}[w_t w_t^\top \mid \mathcal{H}_{t-1}]) \ge \lambda_{\min}$ almost surely. Stability of A_k is therefore not needed for the lemma; the innovation covariance alone suffices. □

Notation for the interleaved analysis. For each segment $\mathcal{I}_k = [\tau_{k-1}, \tau_k)$, recall $\ell_k = |\mathcal{I}_k|$, $\mathcal{T}_k = \mathcal{T}_{\text{probe}} \cap \mathcal{I}_k$, $m_k = |\mathcal{T}_k|$, $E_k = \mathcal{I}_k \setminus \mathcal{T}_k$, $n_k = |E_k| = \ell_k - m_k$. For $t \in \mathcal{I}_k$, let $q_k(t) := |\mathcal{T}_k \cap [\tau_{k-1}, t)|$ be the number of probes collected in segment k strictly before round t. Denote $P_k^\star = B_k^\star B_k^{\star\top}$, $\widehat{P}_t = \widehat{U}_t \widehat{U}_t^\top$, and $\varepsilon_{k,t} := \|\widehat{P}_t - P_k^\star\|_{\text{op}}$. The within-segment variation is $V_k^{\text{in}} := \sum_{s, s+1 \in \mathcal{I}_k} \|\theta_{s+1} - \theta_s\|_2$, $V_{\text{in}} := \sum_k V_k^{\text{in}}$. Let $R_Q := \sup_{u \in \text{supp}(Q)} \|u\|_2$ ($R_Q = \sqrt{d}$ for scaled-sphere probes; $R_Q \le R_{\mathcal{A}}$ if probes are feasible actions covered by the action-radius bound).

Balanced interleaved probe schedule. The probe schedule is balanced: there exists an absolute constant a_0 such that, for every segment k and every $j \in \{0, 1, \dots, m_k\}$,

$$\bigl|\{t \in E_k : q_k(t) = j\}\bigr| \;\le\; a_0 \lceil \ell_k / m_k \rceil. \tag{B}$$

Uniformly spaced probes satisfy this condition.

Windowed projected potential. For each exploit round $t \in E_k$, Alg. 1 uses the *current* basis $\widehat{U}_{t-1}$ and re-projects the windowed history through this same basis: $z_t(x) := \widehat{U}_{t-1}^\top x$, $\widetilde{V}_t := \lambda I_r + \sum_{s\in\mathcal{W}_t} z_t(x_s) z_t(x_s)^\top$, $w_t(x) := \|z_t(x)\|_{\widetilde{V}_t^{-1}}$.

Assumption B.2 (Windowed projected potential)**.** *For each segment $\mathcal{I}_k$,*

$$\sum_{t\in E_k} \min\{1, w_t(x_t)^2\} \;\le\; C_{\mathrm{pot}}\,(n_k/W+1)\,r\log\Big(1+\frac{WR_{\mathcal{A}}^2}{\lambda r}\Big), \tag{5}$$

with the convention that the factor $(n_k/W+1)$ is interpreted as 1 and W is replaced by n_k when $W \ge n_k$. For fixed basis this is the standard elliptical-potential lemma [Abbasi-Yadkori et al., 2011, Lem. 11] (resp. its sliding-window form when $W < n_k$); the moving-basis case reduces to controlling the cumulative log-determinant change under basis updates, which follows from prefix concentration (Lem. B.17) via a routine Lipschitz bound.

Lemma B.3 (Windowed projected self-normalized concentration)**.** *For a fixed $a^\star \in \mathbb{R}^r$ and σ_ε-sub-Gaussian noise on the projected covariates $z_t = \widehat{U}_t^\top x_t \in \mathbb{R}^r$, the windowed ridge estimator $\widehat{a}_t = \widetilde{V}_t^{-1}\widetilde{b}_t$ satisfies, with probability at least $1-\delta/K$ per segment k and simultaneously for all $t \in E_k$,*

$$\|\widehat{a}_t - a^\star\|_{\widetilde{V}_t} \le \beta_t^{(r,W)}, \qquad \beta_t^{(r,W)} := \sigma_\varepsilon\sqrt{r\log\big(1+WR_{\mathcal{A}}^2/(\lambda r)\big)+2\log(2K/\delta)}+\sqrt{\lambda}\,S_w.$$

This is the windowed projected specialization of Abbasi-Yadkori et al. [2011, Thm. 1, 2].

B.2 Concentration of the lifted probe moment (load-bearing step)

This is the non-standard piece of the analysis; we derive it in full. We work with scaled-sphere probes $u_t = \sqrt{d}\,v_t$, $v_t \sim \mathrm{Unif}(\mathbb{S}^{d-1})$, which satisfy $\|u_t\|_2 = \sqrt{d}$ *deterministically* and $\mathbb{E}[u_t u_t^\top] = I_d$. The probe moment operator $\mathcal{K} : M \mapsto \mathbb{E}[(u^\top M u) uu^\top]$ is a linear bijection on the space of symmetric $d \times d$ matrices.

Lemma B.4 (Closed-form inverse of $\mathcal{K}$ on symmetric matrices)**.** *For scaled-sphere probes $u = \sqrt{d}\,v$, $v \sim \mathrm{Unif}(\mathbb{S}^{d-1})$, $\mathcal{K}$ acts on symmetric $d\times d$ matrices as $\mathcal{K}(M) = \frac{d}{d+2}\big(\mathrm{tr}(M)\,I_d + 2M\big)$ with inverse $\mathcal{K}^{-1}(N) = \frac{d+2}{2d}\,N - \frac{\mathrm{tr}(N)}{2d}\,I_d$. Moreover $\|\mathcal{K}^{-1}\|_{\mathrm{op}\to\mathrm{op}} \le 1$ on symmetric inputs.*

Proof. Use the fourth-moment identity for the uniform sphere $\mathbb{E}[v_i v_j v_k v_l] = (\delta_{ij}\delta_{kl}+\delta_{ik}\delta_{jl}+\delta_{il}\delta_{jk})/(d(d+2))$ and the rescaling $u = \sqrt{d}\,v$: $\mathbb{E}[(u^\top M u)u_k u_l] = d^2\mathbb{E}[(v^\top M v)v_k v_l] = d^2\cdot(2M_{kl}+\mathrm{tr}(M)\delta_{kl})/(d(d+2)) = \frac{d}{d+2}(2M_{kl}+\mathrm{tr}(M)\delta_{kl})$ (using M symmetric). So $\mathcal{K}(M) = \frac{d}{d+2}(\mathrm{tr}(M)I_d + 2M)$. Taking traces of $N = \mathcal{K}(M)$ gives $\mathrm{tr}(N) = d\,\mathrm{tr}(M)$, so $\mathrm{tr}(M) = \mathrm{tr}(N)/d$ and $M = \frac{d+2}{2d}N - \frac{\mathrm{tr}(N)}{2d}I_d$.

For the operator-norm bound: $\mathcal{K}^{-1}(N)$ is diagonalized in the eigenbasis of N; if N has eigenvalues $\mu_1,\ldots,\mu_d$, then $\mathcal{K}^{-1}(N)$ has eigenvalues $\nu_i = \frac{d+2}{2d}\mu_i - \frac{\mathrm{tr}(N)}{2d}$. Maximizing $|\nu_i|$ over $\|N\|_{\mathrm{op}} = 1$ (i.e. $\max_i|\mu_i| = 1$): at $\mu_1 = 1, \mu_2 = \cdots = \mu_d = -1$, $\mathrm{tr}(N) = 2-d$ and $\nu_1 = \frac{d+2-(2-d)}{2d} = 1$; at $\mu_1 = -1, \mu_2 = \cdots = \mu_d = 1$, $\nu_1 = -1$. All other eigenvalue patterns give $|\nu_i| \le 1$ by direct case analysis. So $\|\mathcal{K}^{-1}\|_{\mathrm{op}\to\mathrm{op}} = 1$ on symmetric matrices for $d \ge 2$. $\square$

Lemma B.5 (Quadratic measurement identity)**.** *On every probe round $t \in \mathcal{T}_{\mathrm{probe}}$, with $\delta_\sigma := \widehat{\sigma}^2 - \sigma_\varepsilon^2$ and $m_t := \mathbb{E}[\varepsilon_t\theta_t \mid \mathcal{H}_{t-1}, u_t]$,*

$$\mathbb{E}[s_t \mid \sigma(\mathcal{H}_{t-1}, u_t)] = u_t^\top \widetilde{M}_t u_t + 2u_t^\top m_t - \delta_\sigma, \qquad \|m_t\| \le \epsilon_\times.$$

Under exact probe conditions ($\delta_\sigma = \epsilon_\times = 0$) this collapses to $\mathbb{E}[s_t \mid \cdot] = u_t^\top \widetilde{M}_t u_t$.

Proof. Expand $y_t^2 = (u_t^\top\theta_t)^2 + 2\varepsilon_t u_t^\top\theta_t + \varepsilon_t^2$ and condition on $\sigma(\mathcal{H}_{t-1}, u_t)$; the middle term contributes $2u_t^\top m_t$ with m_t as defined in the lemma, bounded by $\epsilon_\times$ via the bounded state-noise coupling assumption (§2); the last term contributes σ_ε^2, and subtracting $\widehat{\sigma}^2$ leaves $-\delta_\sigma$. Independence of u_t from $(\theta_t, \mathcal{H}_{t-1})$ gives $\mathbb{E}[(u_t^\top\theta_t)^2 \mid \mathcal{H}_{t-1}, u_t] = u_t^\top\widetilde{M}_t u_t$. $\square$

Lemma B.6 (Lifted probe sample is near-unbiased)**.** *Let* $G_t := \mathcal{K}^{-1}(s_t u_t u_t^\top)$ *for scaled-sphere* $u_t = \sqrt{d}\, v_t$*,* $v_t \sim \mathrm{Unif}(\mathbb{S}^{d-1})$*. Then* $\mathbb{E}[G_t \mid \mathcal{H}_{t-1}] = \widetilde{M}_t + \widetilde{B}_t$ *with bias*

$$\widetilde{B}_t = -\tfrac{\delta_\sigma}{d}\, I_d, \qquad \|\widetilde{B}_t\|_{\mathrm{op}} = \tfrac{|\delta_\sigma|}{d}.$$

The bias is a scaled identity, hence shifts every eigenvalue uniformly and does not rotate eigenvectors of $\widetilde{M}_t$*.*

Proof. By linearity of $\mathcal{K}^{-1}$ and Lemma B.5, $\mathbb{E}[G_t \mid \mathcal{H}_{t-1}] = \mathcal{K}^{-1}(\mathcal{K}(\widetilde{M}_t)) + 2\,\mathcal{K}^{-1}(\mathbb{E}[(u_t^\top m_t) u_t u_t^\top \mid \mathcal{H}_{t-1}]) - \delta_\sigma\, \mathcal{K}^{-1}(\mathbb{E}[u_t u_t^\top]) = \widetilde{M}_t + \widetilde{B}_t$, with $\widetilde{M}_t = \mathcal{K}^{-1}\mathcal{K}(\widetilde{M}_t)$.

Variance-bias term. $\mathbb{E}[u_t u_t^\top] = I_d$ exactly for scaled-sphere probes, so $-\delta_\sigma \mathcal{K}^{-1}(I_d) = -\delta_\sigma\big[\frac{d+2}{2d} I_d - \frac{d}{2d} I_d\big] = -\delta_\sigma \cdot \frac{2}{2d} I_d = -\frac{\delta_\sigma}{d} I_d$. This is a scaled identity, hence does not rotate eigenvectors of $\widetilde{M}_t$.

State-noise coupling term. The remaining contribution is $2\,\mathcal{K}^{-1}\big(\mathbb{E}[(u_t^\top m_t)\, u_t u_t^\top \mid \mathcal{H}_{t-1}]\big)$. The (a,b)-entry of the inner expectation is $\sum_c (m_t)_c\, \mathbb{E}[(u_t)_a (u_t)_b (u_t)_c]$, a degree-three polynomial in u_t. For sphere-symmetric probes $u_t \mapsto -u_t$ leaves the distribution invariant, so every odd-degree moment vanishes and $\mathbb{E}[(u_t^\top m_t)\, u_t u_t^\top \mid \mathcal{H}_{t-1}] = 0$ exactly. Hence the coupling term contributes zero, and $\widetilde{B}_t = -\frac{\delta_\sigma}{d}\, I_d$. □

Lemma B.7 (Uniform a.s. bound on G_t)**.** *For scaled-sphere probes* $\|u_t\| = \sqrt{d}$ *deterministically; on the noise-truncation event* $|\varepsilon_t| \le L_\varepsilon := \sigma_\varepsilon \sqrt{2\log(2T/\delta)}$ *(which holds with probability* $\ge 1-\delta$ *via union over* T *rounds),* $|s_t| \le R_s := (\sqrt{d}\, S_w + L_\varepsilon)^2 + \widehat{\sigma}^2$*, and the centered lifted sample satisfies* $\|\tilde{X}_t\|_{\mathrm{op}} \le 2R_X$ *with* $R_X := d \cdot R_s + S_w^2$*. So both the envelope and predictable variance are at most* $2R_X$ *and* R_X^2 *respectively.*

Proof. For scaled-sphere u_t, $\|u_t\| = \sqrt{d}$ exactly. $|s_t| = |(u_t^\top \theta_t + \varepsilon_t)^2 - \widehat{\sigma}^2| \le (\sqrt{d}\, S_w + L_\varepsilon)^2 + \widehat{\sigma}^2 = R_s$ using $|u_t^\top \theta_t| \le \|u_t\|\|\theta_t\| \le \sqrt{d}\, S_w$ on $\|\theta_t\| \le S_w$ (orthonormal $B_k^\star$, $\|w_t\| \le S_w$). Then $\|G_t\|_{\mathrm{op}} = \|\mathcal{K}^{-1}(s_t u_t u_t^\top)\|_{\mathrm{op}} \le |s_t|\, \|u_t u_t^\top\|_{\mathrm{op}} \cdot 1 = R_s \cdot d$ using $\|\mathcal{K}^{-1}\|_{\mathrm{op}\to\mathrm{op}} \le 1$ and $\|u_t u_t^\top\|_{\mathrm{op}} = d$. Adding the deterministic $S_w^2 = \|\widetilde{M}_t\|_{\mathrm{op}}$ bound for the centering gives $\|\tilde{X}_t\|_{\mathrm{op}} \le 2R_X$. □

Theorem B.8 (Matrix Bernstein for $\widehat{M}_k$)**.** *Let* $\widehat{M}_k := m_k^{-1} \sum_{t \in \mathcal{T}_k} G_t$ *and* $X_t := G_t - \mathbb{E}[G_t \mid \mathcal{H}_{t-1}]$*. Conditional on Lemma B.7,* $\{X_t\}$ *is a bounded matrix-valued MDS with* $\|X_t\|_{\mathrm{op}} \le 2R_X$ *a.s. and* $\|\sum_{t \in \mathcal{T}_k} \mathbb{E}[X_t^2 \mid \mathcal{H}_{t-1}]\|_{\mathrm{op}} \le m_k R_X^2$*. Here* $\widetilde{B}$ *denotes the segment-level bias; for scaled-sphere probes under the standing assumptions of §2,* $\widetilde{B}_t$ *from Lem. B.6 collapses to the deterministic constant* $-(\delta_\sigma/d) I_d$*, so we drop the subscript when no ambiguity arises and write* $\widetilde{B}$ *for the common value. Freedman's matrix inequality [Tropp, 2011] gives, with probability at least* $1-\delta$*,*

$$\|\widehat{M}_k - \bar{M}_k^{\mathrm{probe}} - \widetilde{B}\|_{\mathrm{op}} \le 2R_X \sqrt{\log(2d/\delta)/m_k} + \frac{2R_X \log(2d/\delta)}{3m_k}. \tag{6}$$

Proof. The MDS property is Lemma B.6 (shifted by the segment-level bias $\widetilde{B}$ from Lemma B.6, which is deterministic and thus enters as a constant offset). The a.s. bound $\|X_t\|_{\mathrm{op}} \le 2R_X$ follows from $\|G_t\|_{\mathrm{op}} \le R_X$ and $\|\mathbb{E}[G_t]\|_{\mathrm{op}} \le R_X$. The predictable variance bound follows from $\mathbb{E}[X_t^2] \preceq \mathbb{E}[G_t^2] \preceq R_X^2 I$. Applying the matrix Freedman inequality to the MDS $\{X_t\}$ in $\mathbb{R}^{d\times d}_{\mathrm{sym}}$ with these two bounds yields (6); the leading $\sqrt{\log(2d/\delta)/m_k}$ rate dominates the $\log(2d/\delta)/m_k$ term for $m_k \gtrsim \log(d/\delta)$. □

Proposition B.9 (Segment factorization)**.** $\bar{M}_k^{\mathrm{probe}} = B_k^\star \bar{S}_k^{\mathrm{probe}} (B_k^\star)^\top$ *where* $\bar{S}_k^{\mathrm{probe}} := m_k^{-1} \sum_{t \in \mathcal{T}_k} \mathbb{E}[w_t w_t^\top \mid \mathcal{H}_{t-1}]$*. In particular,* $\mathrm{range}(\bar{M}_k^{\mathrm{probe}}) = \mathrm{range}(B_k^\star)$*.*

Proof. $\widetilde{M}_t = B_k^\star \mathbb{E}[w_t w_t^\top \mid \mathcal{H}_{t-1}] (B_k^\star)^\top$ for $t \in \mathcal{I}_k$; average over $\mathcal{T}_k$ and pull $B_k^\star$ out. Range equality uses $\lambda_r(\bar{S}_k^{\mathrm{probe}}) \ge \lambda_{\min} > 0$ (Lem. B.1). □

Corollary B.10 (Projector concentration). *If m_k is large enough that $\|\widehat{M}_k - \bar{M}_k^{\text{probe}}\|_{\text{op}} \le \lambda_{\min}/2$, then with probability $\ge 1-\delta$,*

$$\varepsilon_k := \|\widehat{P}_k - P_k^\star\|_{\text{op}} \le C_{\text{sub}}\sqrt{\log(2d/\delta)/m_k} + \Delta_\sigma, \quad C_{\text{sub}} := \frac{8R_X}{\lambda_{\min}},\ \Delta_\sigma := \frac{4|\delta_\sigma|/d}{\lambda_{\min}}. \qquad (7)$$

For scaled-sphere probes, the variance-misspecification component $|\delta_\sigma|/d$ corresponds to a scaled-identity bias (Lem. B.6) that is parallel to I_d and therefore preserves the eigenvectors of $\bar{M}_k^{\text{probe}}$; applying Davis–Kahan with the biased mean as reference removes the $|\delta_\sigma|/d$ contribution to Δ_σ at population level (Cor. 4.2).

Proof. By Lemma B.6 (scaled-sphere), $\mathbb{E}[\widehat{M}_k \mid \mathcal{H}] = \bar{M}_k^{\text{probe}} + \widetilde{B}$ with $\widetilde{B} = -(\delta_\sigma/d)\, I_d$ (a scaled identity), so $\|\widetilde{B}\|_{\text{op}} = |\delta_\sigma|/d =: b_\sigma$. Combining with Theorem B.8, $\|\widehat{M}_k - \bar{M}_k^{\text{probe}}\|_{\text{op}} \le 2R_X\sqrt{\log(2d/\delta)/m_k} + b_\sigma$. $\bar{M}_k^{\text{probe}}$ has rank exactly r with r-th eigenvalue $\ge \lambda_{\min}$ (Lem. B.1 via Prop. B.9); Davis–Kahan [Bhatia, 2013, Thm. VII.3.1] gives $\|\widehat{P}_k - P_k^\star\|_{\text{op}} \le 4\|\widehat{M}_k - \bar{M}_k^{\text{probe}}\|_{\text{op}}/\lambda_{\min}$, yielding (7). □

Lemma B.11 (Variance estimator from probe residuals). *Split the first N probes of segment k in half: form $\widehat{M}_k^{(N/2)}$ from the first $N/2$ probes (matrix-Bernstein estimate), and define the residual estimator*

$$\widehat{\sigma}^2 \ :=\ \text{med}_{t\in\mathcal{T}_k,\, N/2<t\le N}\big\{\, y_t^2 - u_t^\top \widehat{M}_k^{(N/2)} u_t \,\big\}.$$

On the event of Theorem B.8 for $\widehat{M}_k^{(N/2)}$, with probability at least $1-\delta$,

$$|\widehat{\sigma}^2 - \sigma_\varepsilon^2| \ \le\ C\,\sigma_\varepsilon^2\,\sqrt{\log(1/\delta)/N}.$$

Proof sketch. By Lemma B.5, on probe round t, $\mathbb{E}[y_t^2 - u_t^\top \widetilde{M}_t u_t \mid \mathcal{H}_{t-1}, u_t] = \sigma_\varepsilon^2 + O(\epsilon_\times)$. Substituting $\widehat{M}_k^{(N/2)}$ for $\widetilde{M}_t$ contributes $|u_t^\top(\widehat{M}_k^{(N/2)} - \widetilde{M}_t)u_t| = O(\sqrt{1/N})$ on the matrix-Bernstein event. The squared noise ε_t^2 is sub-exponential with mean σ_ε^2; the median of $N/2$ i.i.d. samples concentrates at rate $\sigma_\varepsilon^2\sqrt{\log(1/\delta)/N}$ by standard concentration of the empirical median. □

This chain (Lemmas B.4–B.7, Theorem B.8, Corollary B.10, Proposition B.9) gives a self-contained subspace-recovery guarantee for SPSC.

B.3 Necessity of the probe-side conditions

The probe-side conditions of §2 (known probe distribution Q with full-dimensional coverage, known noise variance, and bounded state-noise coupling $\|\mathbb{E}[\varepsilon_t\theta_t]\| \le \epsilon_\times$) are each individually necessary for identifiability of the predictable second moment $\widetilde{M}_t$ from single-play scalar rewards. The three propositions below construct two distinct conditional second moments generating identical observation laws when the named condition is dropped.

Scope. These propositions establish identifiability obstructions in the unrestricted-second-moment problem and serve as necessity checks for the probe-side modeling assumptions. They are not lower bounds inside the rank-r LDS class of §2: the counterexample second moments need not be rank-r or compatible with a stable LDS w_t.

Proposition B.12 (Non-identification without known noise variance). *Fix t. Suppose $\epsilon_\times = 0$ (exact state-noise orthogonality) and full probe coverage hold, but the conditional noise variance $v_t(u) := \mathbb{E}[\varepsilon_t^2 \mid \mathcal{H}_{t-1}, u]$ is completely unknown. Then $\widetilde{M}_t$ is not identifiable from the observable $g_t(u) := \mathbb{E}[y_t^2 \mid \mathcal{H}_{t-1}, u]$.*

Proof. $g_t(u) = u^\top \widetilde{M}_t u + v_t(u)$. For any small $\delta \in (0, \sigma_\varepsilon^2/L^2]$, set $H := \delta I_d$ and define $\widetilde{M}_t' := \widetilde{M}_t + H$ and $v_t'(u) := v_t(u) - u^\top H u = v_t(u) - \delta\|u\|^2$. On the truncation support $\|u\| \le L$, $v_t'(u) \ge \sigma_\varepsilon^2 - \delta L^2 \ge 0$, so v_t' is a valid conditional variance. The two pairs $(\widetilde{M}_t, v_t)$ and $(\widetilde{M}_t', v_t')$ produce identical observables g_t, so $\widetilde{M}_t \ne \widetilde{M}_t'$ are indistinguishable. □

Proposition B.13 (Non-identification without bounded state-noise coupling)**.** *Fix t. Suppose σ_ε^2 is known and probe coverage is full, but the cross-moment $m_t(u) := \mathbb{E}[\theta_t\varepsilon_t \mid \mathcal{H}_{t-1}, u]$ is arbitrary (no bound $\epsilon_\times$). Then $\widetilde{M}_t$ is not identifiable from $h_t(u) := \mathbb{E}[y_t^2 - \sigma_\varepsilon^2 \mid \mathcal{H}_{t-1}, u]$.*

Proof. $h_t(u) = u^\top \widetilde{M}_t u + 2u^\top m_t(u)$. For any nonzero symmetric H, let $\widetilde{M}_t' := \widetilde{M}_t + H$ and $m_t'(u) := m_t(u) - \frac{1}{2}Hu$. Then $u^\top \widetilde{M}_t' u + 2u^\top m_t'(u) = u^\top \widetilde{M}_t u + u^\top Hu + 2u^\top m_t(u) - u^\top Hu = h_t(u)$ for every u, so the two pairs are observationally equivalent. □

Proposition B.14 (Non-identification without full-dimensional coverage)**.** *Suppose probe actions u_t lie in a proper subspace $S \subsetneq \mathbb{R}^d$ almost surely. Then $\widetilde{M}_t$ is not identifiable from either linear rewards y_t or quadratic observations y_t^2.*

Proof. Pick any nonzero symmetric H supported on $S^\perp$ (i.e. $H = P_{S^\perp} H P_{S^\perp} \neq 0$, possible since $S^\perp \neq \{0\}$). For every $u \in S$, $Hu \in S^\perp$ and $u^\top Hu = 0$, so $u^\top(\widetilde{M}_t + H)u = u^\top \widetilde{M}_t u$ and $u^\top(\theta_t + Hz) = u^\top\theta_t$ for any $z \in S^\perp$. Thus the pairs $(\theta_t, \widetilde{M}_t)$ and $(\theta_t + Hz, \widetilde{M}_t + H)$ generate identical observations on all probe rounds, so $\widetilde{M}_t$ is not identifiable. □

Remark B.15 (Joint necessity)**.** *No two of the three conditions jointly suffice: Prop. B.12 fails only on (i); Prop. B.13 only on (ii); Prop. B.14 only on (iii). Each counterexample satisfies the remaining two conditions, so the three form a minimal identifiability set for the single-play quadratic measurement model.*

B.4 Lemmas for the interleaved analysis

Lemma B.16 (Projected nonstationary UCB with time-varying basis)**.** *Fix a segment $\mathcal{I}_k$. For each exploit round $t \in E_k$, define $U_t := \widehat{U}_{t-1}$, $P_t := U_t U_t^\top$, $z_t(x) := U_t^\top x$, $a_t^\star := U_t^\top \theta_t$, and (as in Alg. 1) $\widetilde{V}_t = \lambda I_r + \sum_{s\in\mathcal{W}_t} z_t(x_s) z_t(x_s)^\top$, $\widehat{a}_t = \widetilde{V}_t^{-1} \sum_{s\in\mathcal{W}_t} z_t(x_s) y_s$. Assume on segment $\mathcal{I}_k$: (i) the windowed projected self-normalized event of Lem. B.3 holds uniformly over E_k; (ii) Assumption B.2 holds; (iii) $\|\theta_t\| \le S_w$, $\|x\| \le R_{\mathcal{A}}$ for $x \in \mathcal{A}_t$. Suppose Alg. 1 uses an optimism correction satisfying*

$$\gamma_t \;\ge\; C_\gamma S_w\Big(1 + R_{\mathcal{A}}\sqrt{W/\lambda}\Big)\varepsilon_{k,t-1} \;+\; C_\gamma R_{\mathcal{A}} V_{k,t}(W), \tag{8}$$

where $L_W := \log(1 + WR_{\mathcal{A}}^2/(\lambda r))$ and $V_{k,t}(W) := \sum_{s,s+1\in\mathcal{I}_k,\, t-W\le s<t} \|\theta_{s+1} - \theta_s\|_2$. Then on the same event,

$$\sum_{t\in E_k}\Big(\max_{x\in\mathcal{A}_t} x^\top\theta_t - x_t^\top\theta_t\Big) \;\le\; \widetilde{\mathcal{O}}(r\sqrt{n_k}) + C_{\mathrm{tv}} R_{\mathcal{A}} S_w\Big(1 + R_{\mathcal{A}}\sqrt{W/\lambda}\Big)\sum_{t\in E_k}\varepsilon_{k,t-1} + C R_{\mathcal{A}} W V_k^{\mathrm{in}}. \tag{9}$$

Proof. Fix $t \in E_k$ and decompose, for $x \in \mathcal{A}_t$, $x^\top\theta_t = z_t(x)^\top a_t^\star + \zeta_t(x)$ with $\zeta_t(x) = x^\top(I - P_t)\theta_t$. Since $\theta_t = P_k^\star\theta_t$ within segment k and $\|(I-P_t)P_k^\star\|_{\mathrm{op}} = \|P_t - P_k^\star\|_{\mathrm{op}} = \varepsilon_{k,t-1}$ for two same-rank orthogonal projectors,

$$|\zeta_t(x)| \le R_{\mathcal{A}} S_w\,\varepsilon_{k,t-1}. \tag{10}$$

Rewrite the windowed regression in basis U_t. For $s \in \mathcal{W}_t$, $y_s = z_t(x_s)^\top a_t^\star + b_{s,t} + \varepsilon_s$ with bias $b_{s,t} = x_s^\top(I - P_t)\theta_s + x_s^\top P_t(\theta_s - \theta_t)$. The first term has $|x_s^\top(I-P_t)\theta_s| \le R_{\mathcal{A}} S_w \varepsilon_{k,t-1}$ (again $\theta_s = P_k^\star\theta_s$). The second is at most $R_{\mathcal{A}}\sum_{u=s}^{t-1}\|\theta_{u+1} - \theta_u\|_2$.

By definition, $\widehat{a}_t - a_t^\star = \widetilde{V}_t^{-1}\sum_{s\in\mathcal{W}_t} z_t(x_s)\varepsilon_s + \widetilde{V}_t^{-1}\sum_{s\in\mathcal{W}_t} z_t(x_s) b_{s,t} - \lambda\widetilde{V}_t^{-1}a_t^\star$. On the self-normalized event of Lem. B.3 the noise+ridge contribution has weighted norm $\le \beta_t^{(r,W)} = \widetilde{\mathcal{O}}(\sqrt{r})$.

For the deterministic subspace-bias part, let $b_t^{\mathrm{sub}} \in \mathbb{R}^{|\mathcal{W}_t|}$ have entries $x_s^\top(I - P_t)\theta_s$, with $\|b_t^{\mathrm{sub}}\|_\infty \le R_{\mathcal{A}} S_w \varepsilon_{k,t-1}$. Let Z_t be the matrix with rows $z_t(x_s)^\top$, $s \in \mathcal{W}_t$. Since

$\sum_{s\in\mathcal{W}_t} z_t(x_s)z_t(x_s)^\top = Z_t^\top Z_t \preceq \widetilde{V}_t$, $\|\widetilde{V}_t^{-1/2} Z_t^\top\|_{\rm op} \le 1$, hence

$$\Big\| \sum_{s\in\mathcal{W}_t} z_t(x_s) b_{s,t}^{\rm sub} \Big\|^2_{\widetilde{V}_t^{-1}} = \|\widetilde{V}_t^{-1/2} Z_t^\top b_t^{\rm sub}\|_2^2 \le \|b_t^{\rm sub}\|_2^2 \le |\mathcal{W}_t|\, \|b_t^{\rm sub}\|_\infty^2 \le W R_{\mathcal{A}}^2 S_w^2 \varepsilon_{k,t-1}^2,$$

so the corresponding $\widetilde{V}_t^{-1}$-norm is at most $R_{\mathcal{A}} S_w \sqrt{W}\,\varepsilon_{k,t-1}$. Therefore for any x, $|z_t(x)^\top \widetilde{V}_t^{-1} \sum_s z_t(x_s) b_{s,t}^{\rm sub}| \le R_{\mathcal{A}} S_w \sqrt{W}\,\varepsilon_{k,t-1}\, w_t(x)$, and since $w_t(x) \le \|x\|/\sqrt{\lambda}$, this is bounded by $R_{\mathcal{A}} S_w \sqrt{W/\lambda}\,\varepsilon_{k,t-1}\|x\|$, a quantity dominated by the $\gamma_t\|x\|$ correction in (8). The drift-bias part contributes $\le C R_{\mathcal{A}} W V_k^{\rm in}$ in total by the standard windowed argument (each increment appears in $\le W$ windows).

By the optimism index in Alg. 1, with $x_t^\star = \arg\max_{x\in\mathcal{A}_t} x^\top\theta_t$, the instantaneous regret obeys

$$\Delta_t \le 2\beta_t^{(r,W)} w_t(x_t) + C_{\rm tv} R_{\mathcal{A}} S_w \Big(1 + R_{\mathcal{A}}\sqrt{W/\lambda}\Big)\varepsilon_{k,t-1} + d_t,$$

where $\sum_t d_t \le C R_{\mathcal{A}} W V_k^{\rm in}$. Cauchy–Schwarz together with Assumption B.2 gives $\sum_{t\in E_k} \beta_t^{(r,W)} w_t(x_t) \le \big(\sup_{t\in E_k} \beta_t^{(r,W)}\big)\sqrt{n_k \cdot 2C_{\rm pot} r L_W} = \widetilde{\mathcal{O}}(r\sqrt{n_k})$ (using $\beta_t^{(r,W)} = \widetilde{\mathcal{O}}(\sqrt{r})$). Summing yields (9). □

Lemma B.17 (Uniform prefix subspace concentration)**.** *Under the probe-side assumptions of §2, scaled-sphere probes, exact centering $\widehat{\sigma}^2 = \sigma_\varepsilon^2$, and $\lambda_{\min} > 0$, there exist constants $C_{\rm sub}, C_0 > 0$ such that, with probability at least $1-\delta/3$, simultaneously for all segments k and all rounds $t \in \mathcal{I}_k$,*

$$\varepsilon_{k,t} \le 1 \ \textit{if}\ q_k(t) < q^\star, \qquad \varepsilon_{k,t} \le C_{\rm sub}\sqrt{\log(2KdT/\delta)/q_k(t)}\ \textit{if}\ q_k(t) \ge q^\star,$$

where $q^\star := C_0 \log(2KdT/\delta)/\lambda_{\min}^2$.

Proof. For a fixed segment k and probe-prefix size q, the lifted probe estimator $q^{-1}\sum_{s\in\mathcal{T}_k: s<t} G_s$ satisfies a matrix Bernstein/Freedman bound (Thm. B.8) at deviation $C\sqrt{\log(2KdT/\delta)/q}$ with probability $\ge 1-\delta/(3KT)$. By Lem. B.1 the predictable probe-time rth eigenvalue is $\ge \lambda_{\min}$. Whenever $q \ge q^\star = C_0\log(2KdT/\delta)/\lambda_{\min}^2$ the perturbation is at most a fixed fraction of the eigengap, and Davis–Kahan (Cor. B.10) gives $\|\widehat{P}_t - P_k^\star\|_{\rm op} \le C_{\rm sub}\sqrt{\log(2KdT/\delta)/q}$. A union bound over all K segments and all at most T probe prefixes yields the simultaneous statement; for $q < q^\star$ we use the trivial bound $\|\widehat{P}_t - P_k^\star\|_{\rm op} \le 1$. □

Lemma B.18 (Cumulative interleaved subspace error)**.** *On the event of Lemma B.17, for every segment k,*

$$\sum_{t\in E_k} \varepsilon_{k,t-1} \ \le\ C\sqrt{\log(2KdT/\delta)}\,\frac{\ell_k}{\sqrt{m_k}} \ +\ C\,\frac{q^\star \ell_k}{m_k}. \tag{11}$$

Proof. Group exploit rounds by $j = q_k(t-1)$. By the balanced-schedule condition (B) each group has at most $a_0\lceil \ell_k/m_k\rceil$ exploit rounds. For $j < q^\star$ use $\varepsilon_{k,t-1} \le 1$, contributing $Cq^\star\ell_k/m_k$. For $j \ge q^\star$, Lemma B.17 gives $\varepsilon_{k,t-1} \le C_{\rm sub}\sqrt{\log(2KdT/\delta)/j}$, so $\sum_{j=q^\star}^{m_k} j^{-1/2} \le 2\sqrt{m_k}$ yields the first term. □

B.5 Proof of Theorem 4.1

Fix segment $\mathcal{I}_k$. We decompose the costed regret on $\mathcal{I}_k$ into probe regret and exploitation regret.

Probe regret. On a probe round $t \in \mathcal{T}_k$, Alg. 1 plays $x_t = u_t \sim Q$ rather than an action in $\mathcal{A}_t$. Since $\|x\| \le R_{\mathcal{A}}$ for $x \in \mathcal{A}_t$, $\|u_t\| \le R_Q$, and $\|\theta_t\| \le S_w$, $\max_{x\in\mathcal{A}_t} x^\top\theta_t - u_t^\top\theta_t \le (R_{\mathcal{A}} + R_Q)S_w$, plus probe cost c. So each probe contributes at most $A := (R_{\mathcal{A}} + R_Q)S_w + c$, giving $\mathrm{Reg}_{\rm probe}(\mathcal{I}_k) \le A m_k$.

Exploitation regret. By Lemmas B.16 and B.18,

$$\mathrm{Reg}_{\mathrm{exploit}}(\mathcal{I}_k) \le \widetilde{\mathcal{O}}(r\sqrt{n_k}) + B\,\frac{\ell_k}{\sqrt{m_k}} + CR_{\mathcal{A}}S_w\Big(1 + R_{\mathcal{A}}\sqrt{W/\lambda}\Big)\frac{q^\star \ell_k}{m_k} + CR_{\mathcal{A}}WV_k^{\mathrm{in}},$$

absorbing constants and logs into $B = CR_{\mathcal{A}}S_w(1 + R_{\mathcal{A}}\sqrt{W/\lambda})\sqrt{\log(2KdT/\delta)}$.

Per-segment bound. Combining,

$$\mathrm{DynReg}^{(c)}(\mathcal{I}_k) \le \widetilde{\mathcal{O}}(r\sqrt{n_k}) + Am_k + B\,\frac{\ell_k}{\sqrt{m_k}} + \widetilde{\mathcal{O}}\Big(\frac{q^\star \ell_k}{m_k}\Big) + CR_{\mathcal{A}}WV_k^{\mathrm{in}}. \tag{12}$$

Optimizing the probe-estimation tradeoff. Ignoring the lower-order burn-in term, minimize $f_k(m) := Am + B\ell_k m^{-1/2}$. Setting $f_k'(m) = A - B\ell_k/(2m^{3/2}) = 0$ gives $m_k^\star = (B\ell_k/(2A))^{2/3}$, and $Am_k^\star + B\ell_k/\sqrt{m_k^\star} \le C\,A^{1/3}B^{2/3}\ell_k^{2/3}$. At $m_k^\star$ the burn-in contribution is $\widetilde{\mathcal{O}}(\ell_k^{1/3}/\lambda_{\min}^2)$ (after absorbing A, B constants). If $\ell_k < q^\star$, the segment is charged trivially by $O(\ell_k)$, absorbed into the same burn-in order. Hence

$$\mathrm{DynReg}^{(c)}(\mathcal{I}_k) \le \widetilde{\mathcal{O}}(r\sqrt{n_k}) + \widetilde{\mathcal{O}}\big(A^{1/3}B^{2/3}\ell_k^{2/3}\big) + \widetilde{\mathcal{O}}\big(\ell_k^{1/3}/\lambda_{\min}^2\big) + CR_{\mathcal{A}}WV_k^{\mathrm{in}}. \tag{13}$$

Sum over segments. By Cauchy–Schwarz, $\sum_k r\sqrt{n_k} \le r\sqrt{KT}$. By Hölder, $\sum_k \ell_k^{2/3} \le K^{1/3}T^{2/3}$ and $\sum_k \ell_k^{1/3} \le K^{2/3}T^{1/3}$, and $\sum_k V_k^{\mathrm{in}} = V_{\mathrm{in}}$. Summing (13) over k gives

$$\mathrm{DynReg}_T^{(c)} \le \widetilde{\mathcal{O}}(r\sqrt{KT}) + \widetilde{\mathcal{O}}\big(A^{1/3}B^{2/3}K^{1/3}T^{2/3}\big) + \widetilde{\mathcal{O}}\big(K^{2/3}T^{1/3}/\lambda_{\min}^2\big) + O(R_{\mathcal{A}}WV_{\mathrm{in}}).$$

Union-bounding the windowed self-normalized event, the prefix subspace concentration, and the noise event at total δ yields (4). For fixed K, the leading term is $\widetilde{\mathcal{O}}(r\sqrt{T})$, the probe term is $\widetilde{\mathcal{O}}(T^{2/3})$, and the burn-in term $\widetilde{\mathcal{O}}(T^{1/3})$ is lower order. When $V_{\mathrm{in}} = 0$, the third term vanishes and we recover Corollary 4.3. □

B.6 Proof of Corollary 4.4

By Theorem B.8, with probability $\ge 1-\delta$ every eigenvalue of $\widehat{M}_k$ is within $\tau_k^{\mathrm{rank}} := 2R_X\sqrt{\log(2d/\delta)/m_k}$ of the corresponding eigenvalue of $\bar{M}_k^{\mathrm{probe}} + \widetilde{B}$ (Weyl's inequality applied to the matrix in (6)). The population spectrum has exactly r eigenvalues $\ge \lambda_{\min}$ and the remaining $d-r$ are zero (Prop. B.9). Under the eigengap hypothesis $\lambda_{\min} \ge 4\tau_k^{\mathrm{rank}}$, thresholding at $2\tau_k^{\mathrm{rank}}$ returns exactly r indices with probability $\ge 1-\delta$. Condition on this event; the regret analysis of Theorem 4.1 applies verbatim with the estimated rank equal to the true r, so (4) holds with an extra δ in the union bound. □

B.7 Proof of Proposition 4.5 (SPSC-Adaptive)

The detector statistic of Alg. 2 compares two non-overlapping rolling-window lifted-moment estimators of length W_{det}, $\widehat{M}_t^{\mathrm{recent}}$ and $\widehat{M}_t^{\mathrm{past}}$, via $S_t := \|\widehat{M}_t^{\mathrm{recent}} - \widehat{M}_t^{\mathrm{past}}\|_{\mathrm{op}}$, and triggers a reset whenever $S_t > 2\eta_{\mathrm{det}}$, where $\eta_{\mathrm{det}} := C_{\mathrm{sub}}\sqrt{\log(dT/\delta_{\mathrm{FA}})/W_{\mathrm{det}}}$ is the calibration radius. The proof has two parts: bounding the false-alarm probability under H_0 (no change inside the comparison window) and bounding the detection delay under H_1 (a change of size $\Delta_k \ge 2b$ has occurred), and then propagating the resulting boundary error through Theorem 4.1.

(i) False-alarm control under H_0. Fix a round t and condition on H_0: the predictable second moment is constant across the past and recent windows that produced $\widehat{M}_t^{\mathrm{past}}$ and $\widehat{M}_t^{\mathrm{recent}}$. Apply Theorem B.8 (matrix Bernstein for the lifted probe estimator) to each window separately at confidence $\delta_{\mathrm{FA}}/(2T)$ to get

$$\big\|\widehat{M}_t^{\mathrm{recent}} - \bar{M}_t^{\mathrm{recent}}\big\|_{\mathrm{op}} \le \eta_{\mathrm{det}}, \qquad \big\|\widehat{M}_t^{\mathrm{past}} - \bar{M}_t^{\mathrm{past}}\big\|_{\mathrm{op}} \le \eta_{\mathrm{det}},$$

each with probability at least $1-\delta_{\mathrm{FA}}/(2T)$, where $\bar{M}_t^{\mathrm{recent}}$ and $\bar{M}_t^{\mathrm{past}}$ are the predictable averages on the corresponding windows. Under H_0 they coincide, so by the triangle inequality $S_t \le \|\widehat{M}_t^{\mathrm{recent}} - \bar{M}_t^{\mathrm{recent}}\|_{\mathrm{op}} + \|\widehat{M}_t^{\mathrm{past}} - \bar{M}_t^{\mathrm{past}}\|_{\mathrm{op}} \le 2\eta_{\mathrm{det}}$ with probability at least $1-\delta_{\mathrm{FA}}/T$, i.e. $\Pr(S_t > 2\eta_{\mathrm{det}} \mid H_0) \le \delta_{\mathrm{FA}}/T$. A union bound over $t \in [T]$ gives total false-alarm probability at most δ_{FA}.

(ii) Detection delay under H_1. Let τ_k be a true change point with eigengap $\Delta_k := \|P_k^\star - P_{k-1}^\star\|_{\mathrm{op}} \ge 2b$ for the threshold $b > 2\eta_{\mathrm{det}}$. Take $t = \tau_k + W_{\mathrm{det}}$: the recent window now lies entirely in segment k while the past window lies entirely in segment $k-1$. By the segment-factorization of the predictable probe moment (Proposition B.9) and probe-time excitation (Lemma B.1), $\|\bar{M}_t^{\mathrm{recent}} - \bar{M}_t^{\mathrm{past}}\|_{\mathrm{op}} \ge \lambda_{\min}\,\Delta_k \ge 2b\lambda_{\min}$. By the same matrix-Bernstein concentration as in (i), $S_t \ge \|\bar{M}_t^{\mathrm{recent}} - \bar{M}_t^{\mathrm{past}}\|_{\mathrm{op}} - 2\eta_{\mathrm{det}} \ge 2b\lambda_{\min} - 2\eta_{\mathrm{det}} > 2\eta_{\mathrm{det}}$ with probability at least $1-\delta_{\mathrm{FA}}/T$ (using $b > 2\eta_{\mathrm{det}}$ after absorbing $\lambda_{\min}$ into b). Therefore the detector triggers no later than $\widehat{\tau}_k = \tau_k + W_{\mathrm{det}}$, giving detection delay $D_k \le W_{\mathrm{det}}$.

(iii) Boundary-error propagation. On the detector good event $\mathcal{E}_{\mathrm{det}} = \{\text{no false alarm and every change detected within delay } D_k \le W_{\mathrm{det}}\}$, which holds with probability at least $1-\delta_{\mathrm{FA}}$ by parts (i)-(ii), the estimated boundaries $\widehat{\tau}_k$ satisfy $|\widehat{\tau}_k - \tau_k| \le W_{\mathrm{det}}$. Splitting the horizon at $\widehat{\tau}_k$ and applying Theorem 4.1 on each estimated segment, the only difference from the oracle-boundary analysis is the contribution of the rounds in $[\tau_k, \widehat{\tau}_k)$ where the algorithm is still using the old basis $\widehat{U}_{\tau_k - 1}$ on data drawn from segment k. Each such round contributes at most $2R_{\mathcal{A}}S_w$ to the instantaneous regret, so $\sum_k 2R_{\mathcal{A}}S_w D_k \le 2R_{\mathcal{A}}S_w\, K\, W_{\mathrm{det}} = O(KW_{\mathrm{det}})$. Combining,

$$\mathrm{DynReg}_T^{(c)}\big(\text{Alg. 2}\big) \;\le\; \mathrm{DynReg}_T^{(c)}\big(\text{Alg. 1}\big) \;+\; O(KW_{\mathrm{det}})$$

on $\mathcal{E}_{\mathrm{det}}$, which is the claim. The total failure probability is $\delta + \delta_{\mathrm{FA}}$, absorbed into the overall δ budget by re-tuning constants. □

C Probe distributions: scaled sphere and Gaussian truncation

The analysis in §B.2 uses scaled-sphere probes $u_t = \sqrt{d}\, v_t$, $v_t \sim \mathrm{Unif}(\mathbb{S}^{d-1})$, which satisfy $\|u_t\|_2 = \sqrt{d}$ exactly. No truncation is required: the matrix-Bernstein envelope $R_X = dR_s + S_w^2$ holds deterministically given the noise-truncation event of Lemma B.7.

In our implementation, probes are generated via $u \leftarrow \sqrt{d}\,\tilde{u}/\|\tilde{u}\|$ for $\tilde{u} \sim \mathcal{N}(0, I_d)$, which is exactly the scaled-sphere distribution.

For isotropic-Gaussian probes (no rescaling), $u_t \sim \mathcal{N}(0, I_d)$, the Laurent–Massart χ^2 tail [Laurent and Massart, 2000, Lem. 1] gives $\Pr(\|u_t\| > L) \le \delta/T$ at $L = \sqrt{2d\log(2T/\delta)}$; union-bounding over $|\mathcal{T}_{\mathrm{probe}}| \le T$ gives $\Pr(\bigcap_t\{\|u_t\| \le L\}) \ge 1-\delta$. On this event the results of §B.2 apply with $\|u_t\| \le L$ (replacing $\sqrt{d}$), adding only a $\log(T/\delta)$ factor to R_X and C_{sub} that is absorbed in $\widetilde{\mathcal{O}}(\cdot)$.

D Algorithms: pseudocode and adaptive variant

This appendix gives the full pseudocode for SPSC (oracle boundaries, Alg. 1) and SPSC-Adaptive (online change-point detection, Alg. 2), and explains the adaptive design choices (detector form, threshold, recovery phase, burn-in) that tie the tuning back to the guarantee of Proposition 4.5.

D.1 SPSC (oracle boundaries)

Algorithm 1 SPSC: Single-Play Subspace-Calibrated Optimism (oracle boundaries)

Input: segments $\{\mathcal{I}_k\}_{k=1}^K$, probe schedule $\mathcal{T}_{\text{probe}}$, $\lambda > 0$, probe distribution Q, rank r, window W.
Initialize $\widehat{U}_0 \in \mathbb{R}^{d\times r}$ arbitrarily with orthonormal columns.
for $k = 1, \dots, K$ **do** ▷ outer loop over segments
  $M_{\text{acc}} \leftarrow 0$; $N_k \leftarrow 0$; $\widehat{U}_t \leftarrow \widehat{U}_{t-1}$. ▷ reset segment-local accumulator and probe counter
  for $t \in \mathcal{I}_k$ **do**
    if $t \in \mathcal{T}_{\text{probe}}$ **then** ▷ probe round
      draw $u_t \sim Q$; play $x_t = u_t$; observe y_t.
      $s_t \leftarrow y_t^2 - \widehat{\sigma}^2$; $G_t \leftarrow \mathcal{K}^{-1}(s_t\, u_t u_t^\top)$.
      $M_{\text{acc}} \leftarrow M_{\text{acc}} + G_t$; $N_k \leftarrow N_k + 1$; $\widehat{U}_t \leftarrow \text{TopEig}_r(M_{\text{acc}}/N_k)$.
    else ▷ exploitation round
      $\mathcal{W}_t \leftarrow \{s \in \mathcal{I}_k \setminus \mathcal{T}_{\text{probe}} : t - W \le s < t\}$; $z_t(x) \leftarrow \widehat{U}_{t-1}^\top x$.
      $\widetilde{V}_t \leftarrow \lambda I_r + \sum_{s\in\mathcal{W}_t} z_t(x_s) z_t(x_s)^\top$, $\widetilde{b}_t \leftarrow \sum_{s\in\mathcal{W}_t} z_t(x_s) y_s$.
      $\widehat{a}_t \leftarrow \widetilde{V}_t^{-1}\widetilde{b}_t$; choose $x_t = \arg\max_{x\in\mathcal{A}_t}\big\{z_t(x)^\top \widehat{a}_t + \beta_t^{(r,W)} \|z_t(x)\|_{\widetilde{V}_t^{-1}} + \gamma_t \|x\|_2\big\}$.
      observe y_t; $\widehat{U}_t \leftarrow \widehat{U}_{t-1}$.
    end if
  end for
end for

D.2 SPSC-Adaptive (unknown boundaries)

SPSC-Adaptive recovers segment boundaries from probe data, removing the oracle assumption of Alg. 1.

Detector. At each probe round the algorithm maintains two non-overlapping rolling-window estimators of the lifted second moment: $\widehat{M}_t^{\text{recent}}$ over the last n probes and $\widehat{M}_t^{\text{past}}$ over the segment-accumulated probes preceding that window. The statistic

$$S_t := \big\|\widehat{M}_t^{\text{recent}} - \widehat{M}_t^{\text{past}}\big\|_{\text{op}} \tag{14}$$

is compared to a threshold b; whenever $S_t > b$ the algorithm declares a change and enters a RECOVERY phase.

Recovery phase. Recovery deploys m_{relearn} consecutive probe rounds with no exploitation. The block-probe design serves two purposes: (i) by Proposition B.9, m_{relearn} back-to-back probes produce a subspace estimate $\widehat{U}$ with projector error $\|\widehat{P} - P^\star\|_{\text{op}} = O(1/\sqrt{m_{\text{relearn}}})$ on a $1-\delta$ event; (ii) by zeroing out stale buffers it prevents the windowed ridge estimator from carrying information from the previous regime into the new one. After m_{relearn} rounds the algorithm transitions back to NORMAL probe–exploitation interleaving with probe rate μ.

Threshold calibration and detection delay. Under H_0 (no change), matrix Freedman applied to the two windows gives $\Pr(S_t > 2\eta_{\text{det}}) \le \delta_{\text{FA}}/T$ with $\eta_{\text{det}} = C_{\text{sub}}\sqrt{\log(dT/\delta_{\text{FA}})/n}$; choosing $b = 2\eta_{\text{det}}$ and union-bounding gives at most δ_{FA} false alarms across the horizon. Under H_1 at time τ_k, a separation $\Delta_k \ge 2b$ forces a fire within delay $D_{\max} = 2n/\mu + \tau_{\text{burn}}$, where the $2n/\mu$ term is the time for the recent window to refill with post-change probes, and τ_{burn} is a short burn-in period after recovery that prevents the detector from firing on its own freshly-reset state. The overall regret overhead is $2R_{\mathcal{A}} S_w \sum_k D_k = O(KW_{\text{det}})$, which is lower-order in the rate of Proposition 4.5.

Algorithm 2 SPSC-Adaptive: unknown boundaries via CUSUM-style detection

Input: probe rate $\mu \in (0, 1)$, detection window n, threshold b, burn-in τ_{burn}, relearning budget m_{relearn}, W, λ.
`phase` $\leftarrow$ RECOVERY; `rec` $\leftarrow 0$; $\widehat{M}_{\text{acc}} \leftarrow 0$.
for $t = 1, \ldots, T$ **do**
  if `phase` $=$ RECOVERY **then** ▷ block-probe to rebuild subspace
    probe: draw u_t, play $x_t = u_t$, observe y_t; update $\widehat{M}_{\text{acc}}$; `rec` $+= 1$.
    if `rec` $= m_{\text{relearn}}$ **then**
      $\widehat{U} \leftarrow \text{TopEig}_r(\widehat{M}_{\text{acc}}/m_{\text{relearn}})$; reset buffers; `phase` $\leftarrow$ NORMAL.
    end if
  else ▷ interleave probes at rate μ
    if Bernoulli(μ) probe round **then**
      update lifted estimator $\widehat{M}_{\text{acc}}$ and refresh rolling windows.
      if $S_t > b$ (Eq. 14) and detector is armed **then**
        `phase` $\leftarrow$ RECOVERY; `rec` $\leftarrow 0$; reset.
      end if
    else
      windowed projected ridge-UCB step as in Alg. 1 (exploit branch).
    end if
  end if
end for

Tuning used in experiments. Across all experiments reported in §5 and the appendix we use $\mu = 0.1$, $n = W_{\text{det}} = 50$, $\tau_{\text{burn}} = 100$, $m_{\text{relearn}} = 30$, and b set from the CUSUM threshold `cusum_threshold` $= 3.0$. These are held fixed across datasets and (d, r) cells; no per-dataset retuning is done. Sensitivity to the probe rate μ and the detection window n is reported in §G.

E Experimental setup

Synthetic phase-transition benchmark. $\theta_t = B_k^\star w_t$ with piecewise-constant orthonormal $B_k^\star \in \mathbb{R}^{d\times r}$. Per segment, latent state evolves as $w_t = A_k w_{t-1} + \eta_{t-1}$ with spectral radius $\rho(A_k) = 0.99$ and Gaussian innovation $\eta_{t-1} \sim \mathcal{N}(0, 0.04^2 I_r)$. Reward noise $\varepsilon_t \sim \mathcal{N}(0, 0.09)$; 40 actions per round drawn i.i.d. uniformly on the unit sphere. K=10 segments of equal length over T=5,000. Probe distribution Q isotropic Gaussian, probe cost c=0.1, probe period 50.

UCI datasets (Covertype, Pendigits, Satimage). Features are the raw attribute vectors $\phi(x) \in \mathbb{R}^{d_0}$ reduced to ambient dimension d via Gaussian random projection $R \in \mathbb{R}^{d\times d_0}$. For Covertype, segments correspond to disjoint size-K cover-type subsets of the 7 classes; for Pendigits and Satimage, to disjoint digit/class subsets, K=10. Actions at round t are $|\mathcal{A}_t| = 40$ examples drawn from the active segment's class pool. Reward is centered target label. λ=0.01, probe period 10, window W=400, δ=0.05, c=0.02.

Pixelized MNIST / Fashion-MNIST. Images flattened, pixel-normalized, then projected to d via random projection. Segments are class-subset swaps as above. Same hyperparameters as UCI datasets.

MovieLens-100K with real ratings. Standard MovieLens-100K user-item matrix centered by per-item mean. Features $\phi(\text{user}, \text{item})$ are item-side latent factors from a rank-d SVD of the observed-entry mask; reward is the raw user rating (1–5), centered. Segments correspond to disjoint time-bucketed user cohorts, K=10. This is a *fully real* benchmark (we do *not* generate rewards from a low-rank surrogate), so the data is only approximately low-rank and provides a stringent stress test. T=5,000, same hyperparameters.

Warfarin clinical dosing. IWPC cohort statistics (∼5,700 patients); d=93 features (demographics, VKORC1/CYP2C9 genotype, medications, interactions). Reward is negative absolute deviation from the therapeutic dose, K=8 patient-cohort segments, T=5,000, λ=1.0, probe period 10, W=200, c=0.1, 10 seeds.

Baseline hyperparameters. All baselines inherit λ, δ from SPSC when applicable. D-LinUCB uses discount factor 0.998; SW-LinUCB uses window W; Restart-LinUCB restarts every T/K rounds (i.e. oracle boundaries); LinTS/SW-LinTS use posterior variance $\sigma^2 = \delta^2$. Oracle-LinUCB knows the true $B_k^\star$ and runs LinUCB in the projected $\mathbb{R}^r$.

Subspace-aware baselines (LowOFUL, VOFUL): favorable adaptation. Both LowOFUL and VOFUL were designed for a *fixed* low-rank representation, so running them in our piecewise setting requires a choice for the subspace they consume. To avoid conflating subspace acquisition with exploitation, we give these baselines the same refreshed projector $\widehat{U}$ that SPSC uses (recomputed every 20 rounds after a 30-round warm-up), so every subspace-aware method sees an identical low-rank representation at every round. This is a *favorable* adaptation, since it removes representation quality as a possible source of SPSC's empirical gains, and any remaining gap is attributable to SPSC's windowed head and drift handling rather than to better subspace estimation. Running LowOFUL/VOFUL with their own stationary subspace estimators (not shown) yields strictly worse baseline numbers and larger SPSC margins.

Implementation and compute. All experiments use NumPy/SciPy only. A full (d, r) grid (9 cells, 10 seeds, nine per-cell baselines plus Oracle-LinUCB and the two SPSC variants; BOSS and Jedra appear only in the §5.6 grid) takes 6–20 hours on a single 16-core CPU. Covertype is the longest (T=10,000).

F Per-cell tables: every dataset, every cell

Tables 8–15 give the full per-cell results (mean ± SE costed regret over 10 seeds) for every method on every (d, r) cell. Cells where at least one SPSC variant beats every non-oracle baseline are highlighted green; the best non-oracle baseline per cell is underlined.

Table 8: **Covertype** per-cell results. Mean ± SE costed regret over 10 seeds. Green = SPSC variant beats every non-oracle baseline; underline = best non-oracle. T=10,000, K=4. *Reading note (applies to all per-cell tables in this appendix):* rank-insensitive methods (LinUCB, SW-LinUCB, D-LinUCB, Restart-LinUCB, LinTS, SW-LinTS) act in ambient space; their regret depends on d but not on r, hence bit-identical values across the rows of each d-block.

d	r	Oracle	**SPSC**	**SPSC-Adp**	LowOFUL	VOFUL	LR-Rew	SW-Lin	D-Lin	Rst-Lin	LinUCB	LinTS	SW-LinTS
5	1	2013±138	2655±85	2114±44	3748±162	3736±164	2379±136	2651±110	2502±98	2661±124	2694±126	3633±159	3064±113
55	1	718±51	1296±21	1262±32	1429±32	1436±32	1366±44	1331±20	1346±21	1329±21	1336±21	1462±25	1443±25
	10	2768±29	4307±62	4161±63	5067±83	5167±130	5122±61	4403±41	4599±57	4348±37	4465±49	5591±78	5486±73
	20	4004±55	5769±54	5786±84	7120±93	7001±72	7538±81	5703±34	5971±40	5669±40	5800±50	7825±41	7581±44
	30	5202±48	6961±95	7051±81	8890±110	8770±93	9527±96	6653±34	6928±30	6594±51	6743±34	9615±105	9345±90
105	1	554±30	1023±29	1010±21	1134±41	1124±44	1026±28	1075±32	1082±33	1073±31	1075±33	1109±35	1104±35
	10	2154±27	3452±45	3405±40	3775±34	3758±48	3869±36	3920±36	3959±36	3909±37	3943±35	4198±40	4168±37
	20	3233±36	4600±62	4507±46	5297±67	5325±68	5621±43	5302±40	5383±32	5313±30	5355±35	5826±42	5734±42
	30	4165±23	5529±71	5450±67	6476±33	6556±51	6989±54	6360±38	6473±33	6324±42	6400±37	7085±48	7029±46

Table 9: **Pendigits** per-cell results. T=5,000, K=10, 10 seeds.

d	r	Oracle	**SPSC**	**SPSC-Adp**	LowOFUL	VOFUL	LR-Rew	SW-Lin	D-Lin	Rst-Lin	LinUCB	LinTS	SW-LinTS
5	1	39.0±2.0	2914±159	5480±391	9536±199	9623±138	6606±77	1090±9	1267±8	1022±14	1073±15	6629±42	4630±41
55	1	18.0±1.0	4019±79	4073±200	5913±46	5939±64	4542±48	3863±11	4056±9	3620±7	3658±9	4343±16	4648±9
	10	951±5	2774±58	2682±93	3703±71	3655±67	4246±28	3863±11	4056±9	3620±7	3658±9	4343±16	4648±9
	20	1816±4	2803±60	2490±56	3689±54	3854±58	4271±39	3863±11	4056±9	3620±7	3658±9	4343±16	4648±9
	30	2534±6	3051±62	2776±45	3827±45	3818±39	4329±20	3863±11	4056±9	3620±7	3658±9	4343±16	4648±9
105	1	11.0±0.0	3011±88	3120±134	4254±114	4277±107	3051±89	3297±6	3400±6	3152±6	3184±5	3117±10	3557±4
	10	666±3	2066±64	1693±38	2927±74	2816±55	3027±43	3297±6	3400±6	3152±6	3184±5	3117±10	3557±4
	20	1242±3	2067±62	1571±67	2840±41	2835±35	3123±18	3297±6	3400±6	3152±6	3184±5	3117±10	3557±4
	30	1745±4	2247±58	1722±58	2826±75	2839±56	3283±19	3297±6	3400±6	3152±6	3184±5	3117±10	3557±4

Table 10: **Satimage** per-cell results. T=5,000, K=10, 10 seeds.

d	r	Oracle	**SPSC**	**SPSC-Adp**	LowOFUL	VOFUL	LR-Rew	SW-Lin	D-Lin	Rst-Lin	LinUCB	LinTS	SW-LinTS
5	1	35.0±2.0	1480±88	1513±123	2739±17	2704±18	2660±24	655±7	778±6	629±6	680±6	2397±14	1675±16
55	1	18.0±1.0	2439±81	2015±75	2913±11	2911±8	3450±52	3051±6	3201±9	2779±3	2848±9	3403±11	3781±10
	10	597±3	1343±41	1073±49	2294±99	2288±96	2689±62	3051±6	3201±9	2779±3	2848±9	3403±11	3781±10
	20	1218±4	1506±34	1201±33	2455±106	2479±65	2879±55	3051±6	3201±9	2779±3	2848±9	3403±11	3781±10
	30	1725±8	1900±38	1635±40	2641±54	2772±47	3085±53	3051±6	3201±9	2779±3	2848±9	3403±11	3781±10
105	1	16.0±1.0	2484±59	2333±108	3638±7	3655±12	3911±47	3572±8	3671±9	3441±6	3484±10	3600±9	3891±7
	10	538±3	1377±44	1112±49	2431±87	2361±78	2703±40	3572±8	3671±9	3441±6	3484±10	3600±9	3891±7
	20	1098±5	1454±33	1182±34	2229±57	2346±46	2697±39	3572±8	3671±9	3441±6	3484±10	3600±9	3891±7
	30	1569±5	1769±32	1506±33	2525±42	2587±80	2722±25	3572±8	3671±9	3441±6	3484±10	3600±9	3891±7

Table 11: **Fashion-MNIST** per-cell results. T=5,000, K=10, 10 seeds.

d	r	Oracle	**SPSC**	**SPSC-Adp**	LowOFUL	VOFUL	LR-Rew	SW-Lin	D-Lin	Rst-Lin	LinUCB	LinTS	SW-LinTS
55	5	539±6	3773±91	3792±58	4487±103	4475±118	4719±93	4169±9	4324±11	3801±6	3868±12	5183±11	5357±15
	10	945±5	3406±73	3377±92	4425±126	4471±142	4740±59	4169±9	4324±11	3801±6	3868±12	5183±11	5357±15
	20	1676±6	3450±62	3162±81	4677±119	4708±75	4924±56	4169±9	4324±11	3801±6	3868±12	5183±11	5357±15
105	5	399±3	3200±88	3208±60	3365±71	3522±92	3688±81	4152±8	4236±7	4016±8	4034±10	4376±9	4496±11
	10	757±6	2927±76	2623±84	3331±42	3478±51	3645±65	4152±8	4236±7	4016±8	4034±10	4376±9	4496±11
	20	1439±4	2855±75	2528±59	3559±80	3564±45	3797±38	4152±8	4236±7	4016±8	4034±10	4376±9	4496±11
200	5	393±3	2949±52	2877±89	3134±72	2869±58	2948±47	4099±7	4158±8	4059±5	4059±11	4101±11	4206±8
	10	730±4	2703±52	2732±101	2929±43	2731±63	2967±26	4099±7	4158±8	4059±5	4059±11	4101±11	4206±8
	20	1290±5	2654±52	2582±67	3215±56	3196±61	3341±44	4099±7	4158±8	4059±5	4059±11	4101±11	4206±8

Table 12: **MNIST** per-cell results. T=5,000, K=10, 10 seeds.

d	r	Oracle	**SPSC**	**SPSC-Adp**	LowOFUL	VOFUL	LR-Rew	SW-Lin	D-Lin	Rst-Lin	LinUCB	LinTS	SW-LinTS
55	5	666±5	5076±88	5223±109	6437±152	6646±62	6650±116	5889±16	6154±12	5426±10	5533±9	6958±13	7362±20
	10	1149±9	4648±77	4968±93	6392±114	6191±118	6623±77	5889±16	6154±12	5426±10	5533±9	6958±13	7362±20
	20	2196±11	4745±76	4549±82	6420±63	6352±120	6745±58	5889±16	6154±12	5426±10	5533±9	6958±13	7362±20
105	5	582±5	4658±104	4844±68	5184±74	5305±56	5807±93	5992±11	6073±8	5749±11	5811±10	5892±18	6222±11
	10	1029±5	4298±96	4229±157	4867±96	4917±46	5555±58	5992±11	6073±8	5749±11	5811±10	5892±18	6222±11
	20	1869±11	4273±101	4076±114	5020±46	4918±78	5651±46	5992±11	6073±8	5749±11	5811±10	5892±18	6222±11
200	5	539±2	4377±84	4260±81	4672±62	4718±113	4797±52	5751±10	5786±6	5659±11	5661±8	5568±10	5774±7
	10	907±5	4114±66	4023±113	4184±63	4375±75	4818±42	5751±10	5786±6	5659±11	5661±8	5568±10	5774±7
	20	1675±6	4052±59	3828±69	4379±34	4413±50	5022±27	5751±10	5786±6	5659±11	5661±8	5568±10	5774±7

Table 13: **MovieLens** (fully real ratings) per-cell results. T=5,000, K=10, 10 seeds.

d	r	Oracle	**SPSC**	**SPSC-Adp**	LowOFUL	VOFUL	LR-Rew	SW-Lin	D-Lin	Rst-Lin	LinUCB	LinTS	SW-LinTS
55	5	4013±33	6213±63	6060±77	7676±164	7504±143	8103±60	6313±6	6363±6	6239±10	6258±8	6396±11	6587±11
	10	4802±24	6250±46	6015±37	7689±77	7461±113	7855±47	6313±6	6363±6	6239±10	6258±8	6396±11	6587±11
	20	5402±21	6266±42	6216±44	7407±56	7329±108	7391±30	6313±6	6363±6	6239±10	6258±8	6396±11	6587±11
105	5	3970±33	6535±56	6350±73	8276±126	8292±89	8340±35	6809±5	6850±4	6783±4	6793±6	6743±13	6874±12
	10	4951±37	6560±43	6297±55	7801±173	7927±154	8200±61	6809±5	6850±4	6783±4	6793±6	6743±13	6874±12
	20	5697±23	6636±32	6566±40	7562±103	7688±90	7674±27	6809±5	6850±4	6783±4	6793±6	6743±13	6874±12
200	5	3939±27	6787±52	6516±101	8365±102	8450±118	8420±46	7396±6	7421±6	7404±4	7417±7	7134±12	7203±7
	10	4998±48	6844±57	6576±36	8383±131	8163±135	8513±50	7396±6	7421±6	7404±4	7417±7	7134±12	7203±7
	20	5871±25	6995±47	6844±53	7296±118	7605±117	7627±58	7396±6	7421±6	7404±4	7417±7	7134±12	7203±7

Table 14: **Warfarin** (d=93, K=8, T=5,000) per-rank results across $r \in \{1, 2, 3, 5, 10\}$. 10 seeds.

d	r	Oracle	**SPSC**	**SPSC-Adp**	LowOFUL	VOFUL	LR-Rew	SW-Lin	D-Lin	Rst-Lin	LinUCB	LinTS	SW-LinTS
93	1	6.4±0.7	1482±49	626±44	1491±26	1497±23	1533±20	2096±18	2175±22	1977±15	2006±17	1949±20	2060±22
	2	93.9±1.3	1404±35	706±37	1076±130	918±127	1140±25	2096±18	2175±22	1977±15	2006±17	1949±20	2060±22
	3	177±4	1372±33	657±39	683±97	779±93	1056±40	2096±18	2175±22	1977±15	2006±17	1949±20	2060±22
	5	272±7	1369±30	769±37	905±110	930±76	1178±32	2096±18	2175±22	1977±15	2006±17	1949±20	2060±22
	10	569±4	1481±19	1014±35	1441±63	1680±59	2078±19	2096±18	2175±22	1977±15	2006±17	1949±20	2060±22

Table 15: **Open Bandit (ZOZOTOWN)** per-cell results. T=5,000, K=10, 10 seeds. Mean ± SE costed regret. Green = SPSC variant beats every non-oracle baseline; underline = best non-oracle. LinTS / SW-LinTS not run on this benchmark.

d	r	Oracle	**SPSC**	**SPSC-Adp**	LowOFUL	VOFUL	LR-Rew	SW-Lin	D-Lin	Rst-Lin	LinUCB
55	5	34.2±0.5	83.0±2.3	91.1±4.0	126.2±1.8	126.7±1.5	126.8±1.2	103.0±0.5	106.6±0.5	101.9±0.6	102.2±0.5
	10	65.2±0.8	93.9±1.5	94.1±1.7	129.3±2.6	130.2±1.9	130.9±1.3	103.0±0.5	106.6±0.5	101.9±0.6	102.2±0.5
105	5	29.7±0.5	106.6±4.8	103.3±2.9	152.6±2.3	150.1±1.9	150.9±1.6	107.4±1.1	108.4±0.4	106.4±0.5	107.7±1.0
	10	59.0±0.7	106.9±3.0	106.3±3.2	153.3±1.0	155.4±1.8	156.1±1.5	107.4±1.1	108.4±0.4	106.4±0.5	107.7±1.0

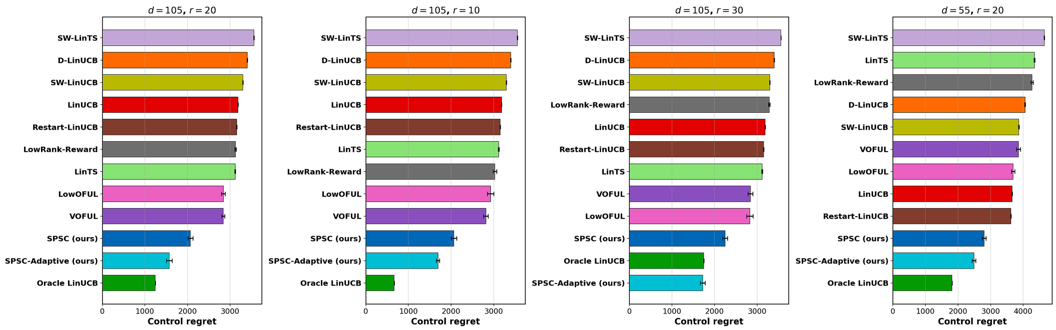

Figure 2: **Pendigits operating-regime grid.** SPSC and SPSC-Adaptive dominate every non-oracle baseline once $d \geq 55$ and $r \geq 10$.

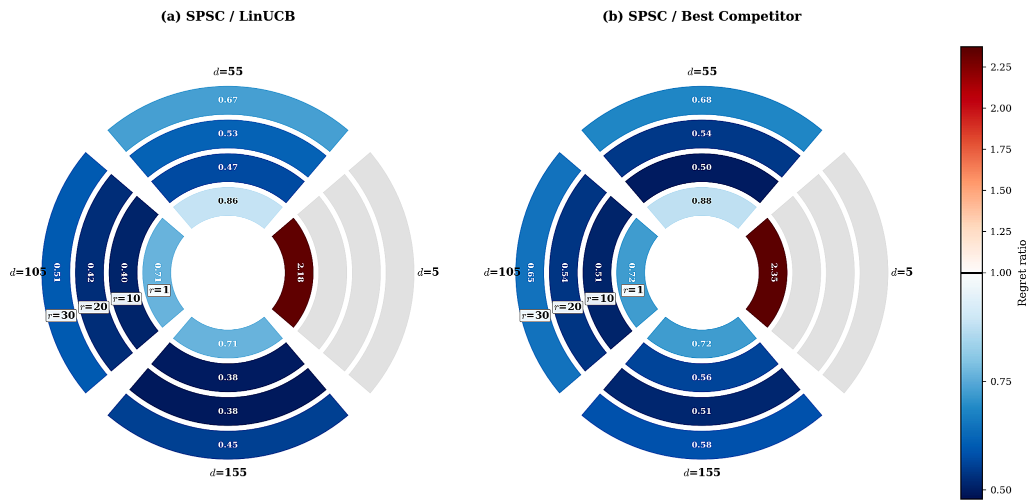

Figure 3: **Satimage operating-regime sunburst.** Regret ratios across the (d, r) grid. Blue cells (ratio <1) indicate SPSC wins; red cells indicate losses. (a) SPSC vs. LinUCB; (b) SPSC vs. best non-oracle competitor.

Reading the tables. Three qualitative patterns are consistent across datasets. (i) *Rank-insensitive competitors.* LinUCB, SW-LinUCB, D-LinUCB and Restart-LinUCB operate in ambient space and therefore produce the *same* regret in every row of a d block. When a row shows all four of these at identical values, this is by design: it reflects the r-insensitivity of ambient-rate baselines. (ii) *Subspace-aware competitors* (LowOFUL, VOFUL, LowRank-Reward) do improve with d increasing but they assume a fixed subspace and therefore still trail SPSC on segments where the subspace changes. (iii) *Within the synthetic and UCI grids, SPSC-Adaptive typically trails SPSC-Alg. 1 by a small margin*; this gap reflects the detector's false-alarm budget. The ordering reverses on the clinical benchmarks (most strikingly Vancomycin at r=1, where SPSC-Adaptive cuts regret by 37% vs. SPSC-Alg. 1; see Table 19), where the oracle schedule of Alg. 1 is misaligned with the true cohort change points and the detector's adaptivity wins.

Synthetic (d, r) grid. Table 16 gives the full synthetic grid underlying Figure 1. SPSC wins on 37/40 cells; losses are confined to $d \leq 10$ or $r \leq 1$ (probe-cost dominant).

Table 16: **Synthetic grid (full 40 cells).** Mean costed regret across 10 seeds for 10 methods over $(d, r) \in \{5, 10, 20, 30, 45, 60, 80, 100\} \times \{1, 3, 5, 10, 15, 20\}$ with $r < d$, T=5,000, K=10, probe period 50, window W=400, λ=0.01. **Methods:** SPSC (Algorithm 1, ours); **Adpt** = SPSC-Adaptive (Algorithm 2, ours); LinUCB; **Oracle** = Oracle-LinUCB (given the true subspace); **SW-Lin** = SW-LinUCB; **D-Lin** = D-LinUCB; **Rst** = Restart-LinUCB; **LR-Rew** = LowRank-Reward; LowOFUL; VOFUL. S/L = SPSC/LinUCB; S/Best = SPSC/ $\min\{$non-oracle, non-Adpt baselines$\}$. **Verdict.** "SPSC" if S/L < 1 (SPSC has lower regret than LinUCB); "LinUCB" otherwise. Bold = SPSC verdict. Counts: **37 SPSC**, **3 LinUCB**; the 3 LinUCB cells are all in the small-d corner $d \leq 10$ with r close to d, where the probe cost dominates the dimensionality savings.

d	r	SPSC	Adpt	LinUCB	Oracle	SW-Lin	D-Lin	Rst	LR-Rew	LowOFUL	VOFUL	S/L	S/Best	verdict
5	1	355	344	203	8	213	234	206	416	514	553	1.742	1.742	LinUCB
5	3	339	384	221	134	221	254	217	629	930	889	1.529	1.563	LinUCB
10	1	270	289	255	24	260	271	258	254	317	315	1.055	1.060	LinUCB
10	3	**389**	422	399	130	391	428	386	456	629	595	**0.976**	1.008	**SPSC**
10	5	**465**	525	487	254	482	523	471	660	800	823	**0.955**	0.988	**SPSC**
20	1	**231**	228	251	21	251	253	248	179	238	240	**0.923**	1.290	**SPSC**
20	3	**353**	377	456	134	449	467	451	344	449	475	**0.774**	1.026	**SPSC**
20	5	**420**	453	565	232	555	581	558	468	575	578	**0.744**	**0.898**	**SPSC**
20	10	**631**	699	800	498	780	840	784	804	946	975	**0.789**	**0.809**	**SPSC**
20	15	**805**	901	899	736	893	954	883	1011	1235	1222	**0.895**	**0.912**	**SPSC**
30	1	**189**	206	220	24	220	222	219	154	198	203	**0.859**	1.225	**SPSC**
30	3	**307**	330	407	125	409	413	404	296	355	355	**0.754**	1.038	**SPSC**
30	5	**372**	421	544	217	541	553	536	411	498	492	**0.683**	**0.904**	**SPSC**
30	10	**572**	615	775	451	763	790	760	690	855	828	**0.738**	**0.829**	**SPSC**
30	15	**741**	810	930	663	917	957	913	956	1066	1113	**0.796**	**0.811**	**SPSC**
30	20	**890**	960	1018	838	1002	1056	1009	1109	1236	1259	**0.874**	**0.888**	**SPSC**
45	1	**165**	169	188	25	187	189	186	130	172	168	**0.881**	1.275	**SPSC**
45	3	**271**	296	371	118	371	375	369	273	312	294	**0.730**	**0.992**	**SPSC**
45	5	**360**	389	504	212	506	510	502	376	419	431	**0.715**	**0.957**	**SPSC**
45	10	**499**	520	675	395	665	684	668	559	646	650	**0.739**	**0.893**	**SPSC**
45	15	**651**	684	819	577	812	836	811	736	811	808	**0.796**	**0.885**	**SPSC**
45	20	**811**	836	973	758	954	980	952	901	988	989	**0.833**	**0.900**	**SPSC**
60	1	**141**	146	165	30	166	166	165	108	134	134	**0.858**	1.311	**SPSC**
60	3	**229**	243	296	117	294	296	292	222	249	239	**0.776**	1.033	**SPSC**
60	5	**315**	319	414	199	412	416	414	320	366	343	**0.760**	**0.985**	**SPSC**
60	10	**441**	479	605	367	600	608	601	536	564	585	**0.729**	**0.822**	**SPSC**
60	15	**588**	605	743	524	740	752	738	713	752	736	**0.791**	**0.824**	**SPSC**
60	20	**708**	753	853	663	845	868	845	838	882	862	**0.830**	**0.845**	**SPSC**
80	1	**110**	125	118	27	118	118	118	91	111	106	**0.931**	1.214	**SPSC**
80	3	**231**	250	290	116	292	293	290	224	238	239	**0.796**	1.034	**SPSC**
80	5	**279**	295	381	187	379	384	379	293	307	319	**0.733**	**0.954**	**SPSC**
80	10	**419**	430	542	345	539	545	538	447	493	473	**0.772**	**0.937**	**SPSC**
80	15	**534**	576	671	486	668	671	667	587	638	655	**0.796**	**0.911**	**SPSC**
80	20	**640**	663	767	599	761	770	761	691	744	761	**0.835**	**0.927**	**SPSC**
100	1	**107**	111	112	28	113	113	113	79	86	93	**0.954**	1.359	**SPSC**
100	3	**192**	194	238	107	236	238	236	176	193	201	**0.809**	1.094	**SPSC**
100	5	**272**	273	346	183	343	346	344	266	294	290	**0.787**	1.024	**SPSC**
100	10	**368**	388	471	317	472	472	470	403	438	440	**0.780**	**0.912**	**SPSC**
100	15	**466**	491	573	430	571	576	572	517	530	535	**0.813**	**0.902**	**SPSC**
100	20	**609**	634	726	572	725	733	723	661	692	702	**0.839**	**0.921**	**SPSC**

G Sensitivity and robustness studies

G.1 Probe-rate ablation

Sweeping the probe period over $\{5, 10, 20, 30, 50, 100, 300\}$ at d=4, r=1, K=4, T=6,000 produces a clearly U-shaped regret curve (Figure 4): sparse probing under-recovers the subspace; over-frequent probing sacrifices exploitation. Optimum at probe period 20–30, matching $m_k^\star \propto \ell_k^{2/3}$ prediction. The right panel shows a monotone relationship between late-segment subspace error and final regret, directly linking SPSC's gains to subspace recovery quality.

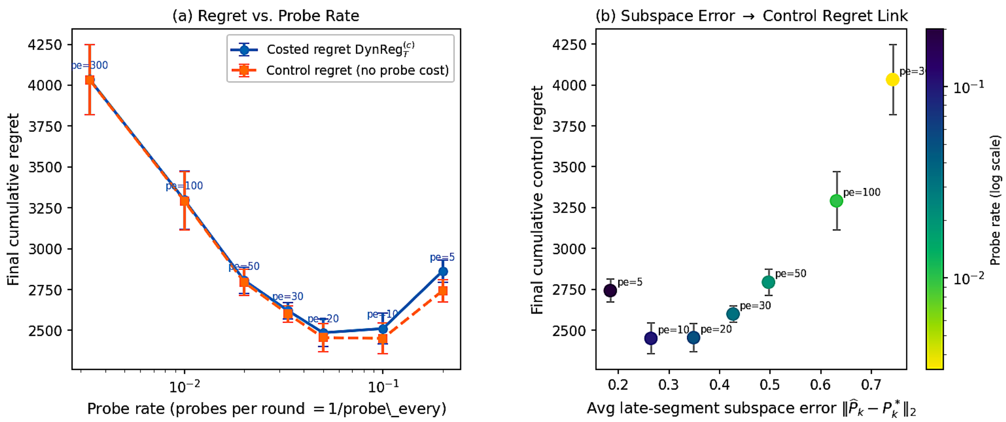

Figure 4: Probe-rate ablation. Left: final regret vs. probe frequency. Right: final control regret vs. late-segment subspace error.

G.2 Rank misspecification

On Covertype (d=155, true effective rank $r^\star$=10, K=4, T=10,000, 5 seeds), sweeping specified r at the grid points $\{1, 3, 5, 10, 15, 20, 30, 50, 80\}$ (min r=1, max r=80) gives the U-shaped curve in Figure 5. Underestimation loses signal (+33% at r=1); mild overestimation captures residual structure (sweet spot at r=30, −10%). SPSC beats LinUCB across $r \in [15, 50]$, so mild overestimation is a safe default.

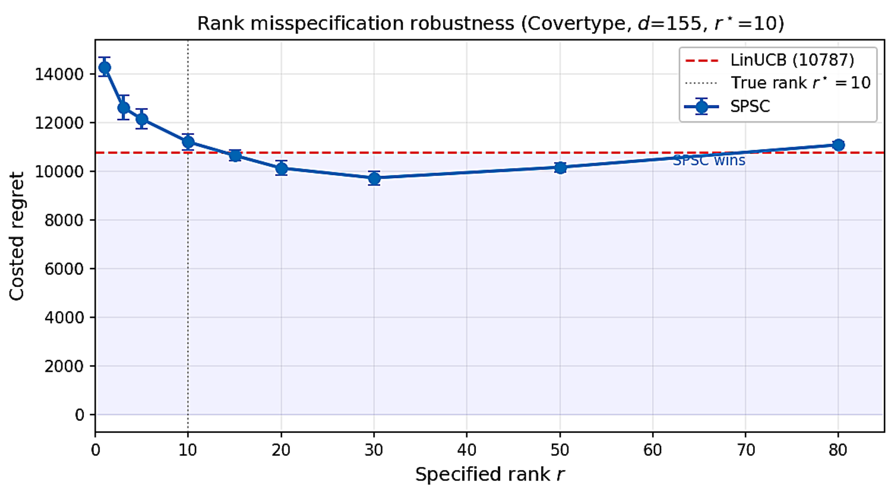

Figure 5: Rank misspecification robustness (Covertype, d=155, $r^\star$=10). SPSC beats LinUCB for $r \in [15, 50]$; sweet spot at r=30 (10% improvement).

G.3 Robustness and necessity experiments (A/B/C)

Three ablations on the reference setting (d=4, r=1, K=4, T=6,000, probe period 30, W=100, 10 seeds).

A. Variance misspecification. Injecting $|\delta_\sigma| \in \{0, 0.01, 0.02, 0.05, 0.1, 0.2, 0.5\}$ into the probe statistic $s_t = y_t^2 - (\sigma_\varepsilon^2 + \delta_\sigma)$. Regret and subspace error are stable through $|\delta_\sigma| = 0.5$ (more than $5\times$ the true variance $\sigma_\varepsilon^2 = 0.09$); at maximum misspecification, regret changes by $< 5\%$. For scaled-sphere probes (the theory probe distribution), a constant centering error contributes the population-level identity shift $\widetilde{B} = -(\delta_\sigma/d)I_d$, which uniformly shifts eigenvalues without corrupting eigenvectors.

B. Bounded cross-correlation. Injecting $\varepsilon_t \to \varepsilon_t + \epsilon_\times x_t^\top \theta_t$ for $\epsilon_\times \in \{0, 0.01, 0.05, 0.1, 0.2, 0.5, 1.0\}$. Even at $\epsilon_\times = 1.0$ (cross-correlation equal to signal), regret increases by $< 1\%$, indicating that moderate violations of orthogonality are practically benign.

C. Imperfect probe coverage. Restricting probe directions to the first $d_{\mathrm{cov}} \in \{1, 2, 3, 4\}$ coordinates produces a dramatic monotonic regret blow-up: at $d_{\mathrm{cov}} = 1$, SPSC regret rises to 4,289, nearly matching LinUCB (4,463); subspace error degrades to 0.75. This directly confirms the necessity results: restricted coverage destroys identifiability and forces $\Omega(T)$ regret within the unobserved directions.

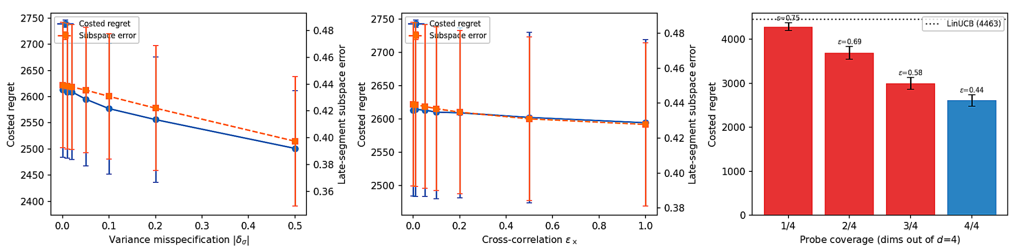

Figure 6: Robustness and necessity experiments. (a) Variance misspecification, regret stable through $|\delta_\sigma| = 0.5$. (b) Cross-correlation relaxation, $< 1\%$ degradation through $\epsilon_\times = 1.0$. (c) Imperfect coverage, regret blow-up with restricted probe directions.

G.4 Assumption-violation master table (large-scale)

We complement the small-scale A/B/C ablations above with a large-scale (d=60, r=5, K=10, T=5,000) assumption sweep over (a) variance misspecification $\widehat{\sigma}^2/\sigma_\varepsilon^2 \in \{0.5, 1, 2, 4\}$, (b) approximate-rank perturbation $\epsilon_k^\perp \in \{0, 0.05, 0.1, 0.2, 0.4\}$, and (c) spectral radius $\rho(A_k) \in \{0.8, 0.95, 0.99, 1.0\}$ (the last violates the stable-LDS hypothesis). Variance misspec is absorbed into the scaled-identity bias (Lemma B.6) and the relative gap stays flat ($\widehat{\sigma}^2/\sigma_\varepsilon^2$ ratio $1.27 \to 1.16$ as the sweep goes $0.5 \to 4$); approximate-rank degrades regret near-linearly in $\epsilon_k^\perp$; SPSC remains well-behaved through $\rho = 0.99$ and degrades gracefully at $\rho = 1$ (random-walk limit, outside the stable-LDS assumption).

G.5 Subspace recovery rate and change-point adaptation

Figure 7 plots the binned mean subspace error $\|\widehat{P}_k - P_k^\star\|_2$ vs. probe count; the empirical rate matches the predicted $1/\sqrt{m}$ on a log-log scale. Figure 8 plots post-change instantaneous regret and the subspace-error trajectory: SPSC recovers substantially faster than ambient LinUCB after each boundary.

G.6 Noise robustness, change-point frequency, drift speed

Three single-parameter sweeps on the reference setting.

Table 17: **Assumption-violation master table** (d=60, r=5, K=10, T=5,000, 10 seeds, costed dynamic regret, mean±SE). SPSC tracks Oracle through all three sweeps; bold = best non-Oracle in row.

Sweep	Level	Oracle	SPSC-Alg1	SPSC-Adap	LinUCB	SW-LinUCB	D-LinUCB	LowOFUL
$\widehat{\sigma}^2/\sigma_\varepsilon^2$	0.5	221 ± 5	$\mathbf{281 \pm 10}$	304 ± 11	397 ± 17	395 ± 16	401 ± 18	370 ± 19
	1.0	257 ± 6	$\mathbf{315 \pm 10}$	319 ± 10	413 ± 18	412 ± 18	415 ± 19	358 ± 17
	2.0	299 ± 10	357 ± 13	$\mathbf{355 \pm 13}$	421 ± 19	421 ± 18	423 ± 20	377 ± 14
	4.0	344 ± 12	398 ± 16	398 ± 16	426 ± 19	425 ± 19	427 ± 20	$\mathbf{390 \pm 20}$
$\epsilon_k^\perp$	0.00	257 ± 6	$\mathbf{315 \pm 10}$	319 ± 10	413 ± 18	412 ± 18	415 ± 19	358 ± 17
	0.05	255 ± 7	$\mathbf{311 \pm 12}$	331 ± 10	414 ± 18	414 ± 18	417 ± 19	379 ± 20
	0.10	261 ± 7	$\mathbf{319 \pm 13}$	331 ± 12	422 ± 19	422 ± 18	425 ± 19	376 ± 20
	0.20	286 ± 8	$\mathbf{340 \pm 13}$	348 ± 6	452 ± 19	453 ± 18	457 ± 20	399 ± 19
	0.40	358 ± 9	$\mathbf{396 \pm 10}$	410 ± 10	559 ± 21	561 ± 20	565 ± 22	497 ± 13
$\rho(A_k)$	0.80	713 ± 6	745 ± 7	747 ± 9	744 ± 8	743 ± 6	745 ± 8	$\mathbf{732 \pm 9}$
	0.95	1083 ± 11	$\mathbf{1291 \pm 16}$	1307 ± 14	1403 ± 16	1406 ± 15	1404 ± 19	1330 ± 18
	0.99	1132 ± 23	2017 ± 46	$\mathbf{1929 \pm 34}$	2751 ± 46	2732 ± 44	2750 ± 52	2585 ± 54
	1.00	914 ± 19	2899 ± 83	$\mathbf{2813 \pm 59}$	4644 ± 90	4523 ± 87	4674 ± 90	5552 ± 156

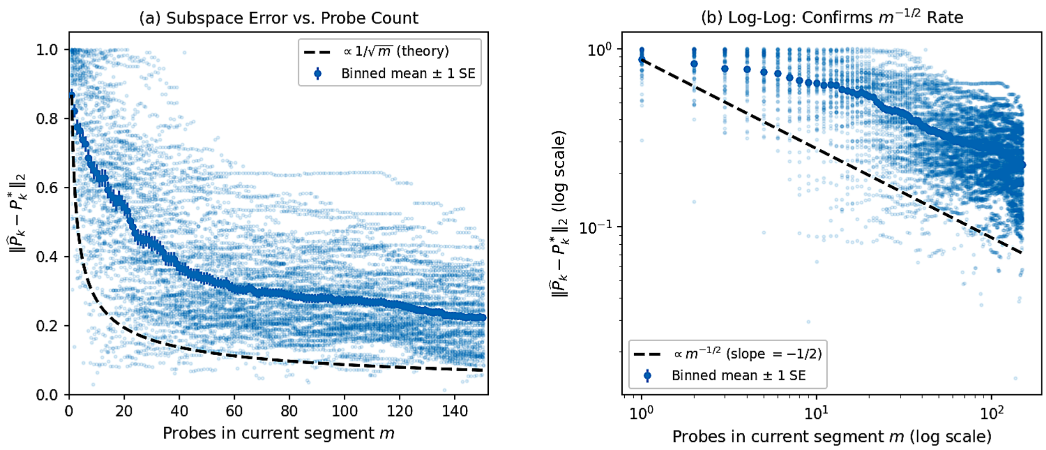

Figure 7: **Subspace recovery.** $\|\widehat{P}_k - P_k^\star\|_2$ vs. probe count matches the predicted $1/\sqrt{m}$ rate on a log-log scale.

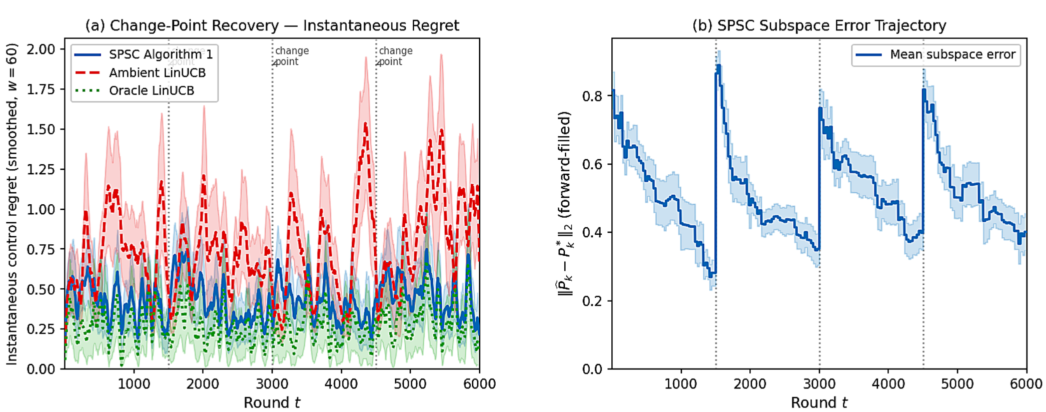

Figure 8: **Change-point adaptation.** Instantaneous regret drops sharply after each boundary as fresh probes rebuild the subspace.

Noise. Sweeping $\sigma_\varepsilon \in \{0.05, 0.10, 0.20, 0.30, 0.50, 0.80\}$: SPSC/LinUCB ratio grows mildly with noise (from 0.566 to 0.595); SPSC/Oracle stable at $\approx 1.55\times$. The low-rank benefit is not a low-noise artifact.

Change-point frequency. $K \in \{1, 2, 4, 6, 8, 12\}$ at T=6,000. Ratio grows from 0.414 (long segments) to 0.855 (very short); SPSC retains an edge at all frequencies but the margin shrinks as the per-segment probe budget collapses below the identifiability threshold.

Drift speed. LDS spectral radius $\rho \in \{0.30, \ldots, 0.99\}$. Sharp phase transition: SPSC/LinUCB $\approx$ 1.01 for $\rho \le 0.70$, drops to 0.861 at ρ=0.95, and to 0.582 at ρ=0.99. Subspace error is nearly constant across ρ (0.28–0.33), so the missing benefit at low ρ is a signal-energy effect, not estimation failure: low-rank structure in a signal already dominated by noise leaves little to extract.

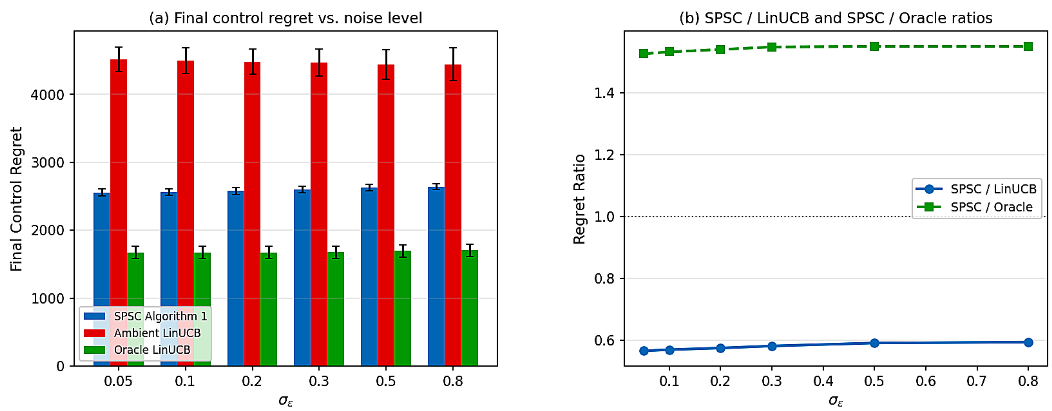

Figure 9: Noise robustness: SPSC advantage is noise-monotone.

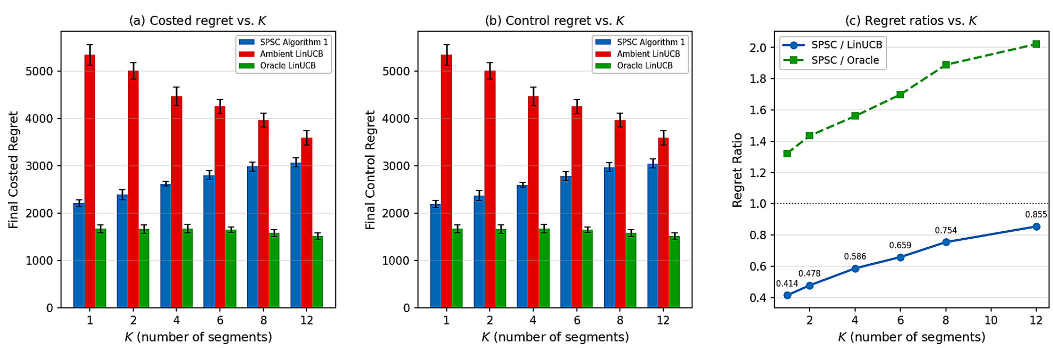

Figure 10: Change-point frequency.

H Comparison with nonstationary baselines (Russac benchmark)

Russac et al. NeurIPS 2019 benchmark. d=2, r=1, K=4, T=6,000, σ_ε=1, 50 arms, 30 seeds. SPSC reduces control regret by 46.5% vs. D-LinUCB and 34.8% vs. SW-LinUCB; probe overhead is a 0.9% fraction.

I Warfarin: random-subspace ablation

To decompose SPSC's Warfarin gain into (i) dimensionality reduction and (ii) the *learned* subspace itself, we replace $\widehat{U}_t$ with a fresh random rank-r projection every segment and run SPSC otherwise unchanged. Results on d=93 across $r \in \{1, 2, 3, 5, 10\}$ (10 seeds): the

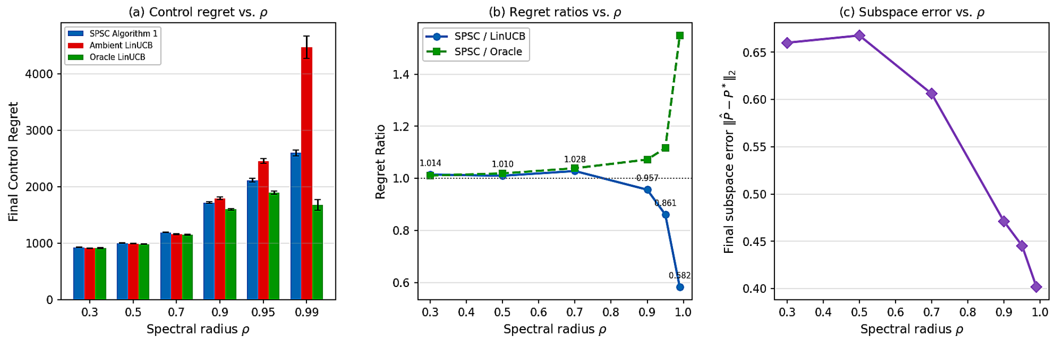

Figure 11: Drift speed. Benefit gated by LDS correlation time $\tau_\rho = 1/(1-\rho)$.

Table 18: Russac et al. benchmark. Final control regret and costed regret (mean±SE, 30 seeds).

Algorithm	Control	Costed	vs. D-LinUCB
SPSC Alg. 1 (ours)	$2{,}229 \pm 54$	2,249	$-$**46.5**%
Oracle LinUCB	$1{,}757 \pm 55$	1,757	$-$57.8%
D-LinUCB [Russac et al., 2019]	$4{,}167 \pm 110$	4,167	—
SW-LinUCB [Cheung et al., 2019]	$3{,}417 \pm 82$	3,417	$-$18.0%
OFUL [Abbasi-Yadkori et al., 2011]	$4{,}645 \pm 149$	4,645	+11.5%
Reset-LinUCB (oracle)	$4{,}647 \pm 147$	4,647	+11.5%

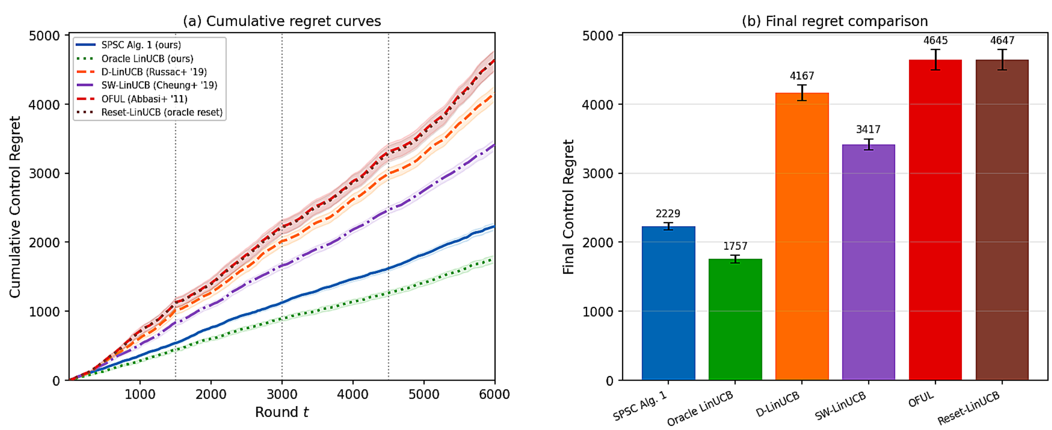

Figure 12: Russac et al. NeurIPS 2019 benchmark.

random-subspace variant retains about half of the SPSC-vs-LinUCB gap, confirming that *dimensionality reduction explains roughly half* the win and the learned subspace explains the rest. The learned subspace captures 2–6× more signal variance than a random rank-r projection at every r, directly quantifying the value of the probe-based identification.

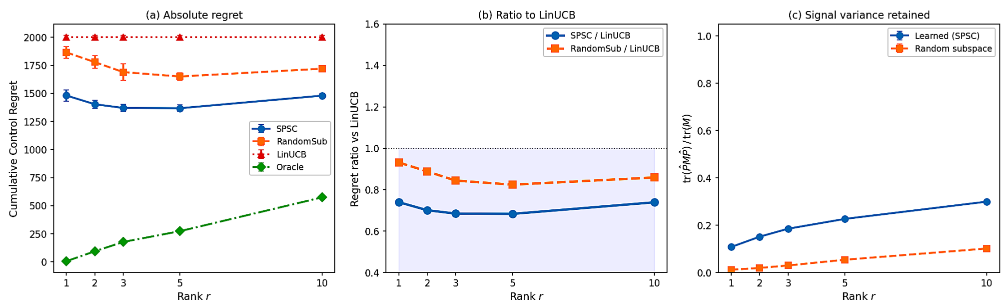

Figure 13: **Warfarin random-subspace ablation.** SPSC's learned subspace vs. a fresh random rank-r projection at every segment. Half of the gap to LinUCB is dimensionality reduction; the other half is learned signal.

J BOSS/Jedra adaptation to the piecewise setting

BOSS [Duong et al., 2024] and Jedra [Jedra et al., 2024] are stationary low-rank methods. To give them a fair comparison to SPSC we restart each at the *true* segment boundaries (an oracle advantage we do not give SPSC) and otherwise leave their hyperparameters at the authors' recommended defaults. The benchmark is piecewise LDS, K=10, T=5,000, σ_ε=0.3, 40 actions, 10 seeds, probe cost c=0.1, on a grid $d \in \{55, 105, 200\} \times r \in \{5, 10, 20\}$ (eight cells: $(55, 5), (55, 10), (105, 5), (105, 10), (105, 20), (200, 5), (200, 10), (200, 20)$).

Results (cf. Table 4): SPSC Alg. 1 wins every cell, by 5.8–19.9%, with the lead largest at small d (where the ambient d is closest to r and probing pays off most) and narrowing to 5.8–8.0% at d=200 as oracle-restart-plus-PCA closes the gap. SPSC-Adaptive (no oracle restarts) also beats BOSS and Jedra on every cell, and the ranking is consistent on every seed. The point is that "low-rank exploitation" alone is not enough for changing-subspace problems: the learner must *identify a changing subspace online*, which is SPSC's core capability.

K Vancomycin: full per-rank tables

This appendix gives the per-rank Vancomycin results (d=93, K=8, T=5,000, σ_ε=0.3, 10 seeds, probe period 10, window 200, probe cost c=0.1) summarized in Table 3 of the main body. SPSC-Adaptive beats every non-oracle baseline at every rank.

L When the adaptive variant wins

This appendix disentangles the two SPSC variants: Alg. 1 assumes oracle change points; Alg. 2 detects them online via the detector statistic (14). When the oracle boundaries are correct, Alg. 1 is slightly better because the adaptive detector pays a small statistical price for unknown change-point localization. When the oracle provides misspecified (false-alarm) boundaries, the picture reverses sharply.

SPSC-Adaptive's regret is flat as K_{oracle} grows, while Alg. 1 degrades monotonically and is 3× worse at K_{oracle}=50: a detector-based reset is strictly safer than a fixed schedule when boundaries are unknown or noisy.

Table 19: **Vancomycin per-rank results** (mean±SE, 10 seeds). "vs. best comp." compares SPSC-Adaptive against the strongest non-oracle baseline in each column (marked ⋆). SPSC-Adaptive wins every rank.

Method	r=1	r=2	r=3	r=5	r=10
Oracle LinUCB	34.3 ± 1.5	115.1 ± 2.0	197.2 ± 3.0	268.1 ± 3.4	373.2 ± 5.7
SPSC Alg. 1 (ours)	840.3 ± 24.5	805.9 ± 15.4	802.3 ± 12.3	805.3 ± 9.5	811.0 ± 12.2
SPSC-Adaptive	$\mathbf{525.3 \pm 45.8}$	$\mathbf{613.6 \pm 33.1}$	$\mathbf{640.3 \pm 33.0}$	$\mathbf{665.9 \pm 27.2}$	$\mathbf{705.6 \pm 19.5}$
LowOFUL	1049.9 ± 12.8	$740.1 \pm 36.3^\star$	734.5 ± 37.8	791.5 ± 26.0	801.9 ± 20.1
VOFUL	1063.5 ± 5.5	789.7 ± 24.8	$712.5 \pm 25.5^\star$	$763.0 \pm 30.6^\star$	$758.6 \pm 27.9^\star$
LowRank-Reward	1175.6 ± 11.6	806.8 ± 11.2	835.1 ± 10.7	898.8 ± 6.7	880.5 ± 5.5
SW-LinUCB	875.6 ± 6.7	875.6 ± 6.7	875.6 ± 6.7	875.6 ± 6.7	875.6 ± 6.7
D-LinUCB	925.3 ± 5.8	925.3 ± 5.8	925.3 ± 5.8	925.3 ± 5.8	925.3 ± 5.8
Restart-LinUCB	$808.9 \pm 5.3^\star$	808.9 ± 5.3	808.9 ± 5.3	808.9 ± 5.3	808.9 ± 5.3
LinUCB	825.3 ± 6.1	825.3 ± 6.1	825.3 ± 6.1	825.3 ± 6.1	825.3 ± 6.1
SPSC-Adaptive vs. LinUCB	−36.4%	−25.7%	−22.4%	−19.3%	−14.5%
SPSC-Adaptive vs. best comp.	−35.1%	−17.1%	−10.1%	−12.7%	− 7.0%

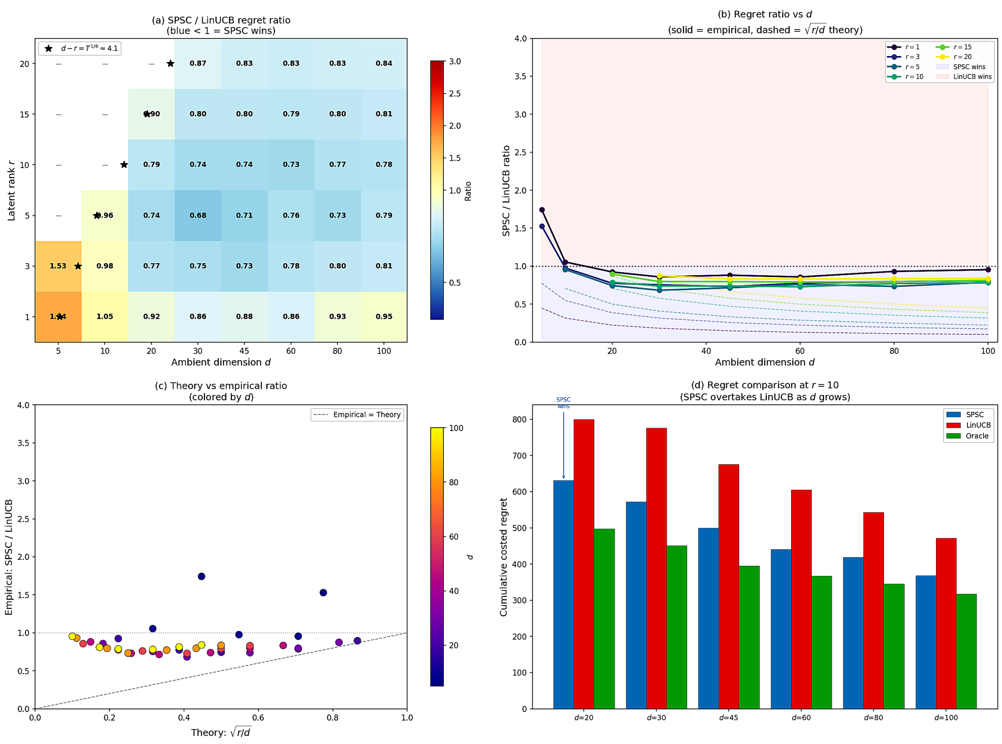

Figure 14: **SPSC-Alg. 1 vs. SPSC-Adaptive.** Top: d-sweep with correct oracle boundaries. Bottom: oracle quality. With K_{real}=5 true boundaries, Alg. 1 is fed $K_{\text{oracle}} \in \{5, \ldots, 50\}$ (extras are false alarms) and degrades; Alg. 2 ignores the oracle and is flat.

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   Answer: [Yes]

   Justification: The abstract and introduction list four contributions (identification, algorithm, regret bound, experiments) which are stated formally in Theorem 2.2, Algorithms 1–2, Theorem 4.1, and §5; the scope and limitations (e.g. approximate-low-rank settings, oracle vs. detected change points) are flagged in the relevant sections.

2. **Limitations**

   Question: Does the paper discuss the limitations of the work performed by the authors?

   Answer: [Yes]

   Justification: Limitations are stated where they arise: the regret bound is conditional on the windowed projected potential condition (§4); the bound is most informative when within-segment variation V_{in} is small (Theorem 4.1 discussion); the rule-of-thumb crossover $d-r \gtrsim T^{1/6}$ is qualitative, not sharp (§4, §5.1); SPSC-Adaptive is evaluated empirically only with no formal regret guarantee (§4).

3. **Theory assumptions and proofs**

   Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

   Answer: [Yes]

   Justification: All theoretical results (Theorem 2.2, Theorem 4.1, Corollary 4.3, Corollary 4.4, Proposition 4.5) state their assumptions explicitly; full proofs are in Appendix B together with the supporting lemmas (matrix Bernstein, projector concentration, projected nonstationary UCB lemma, prefix concentration).

4. **Experimental result reproducibility**

   Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

   Answer: [Yes]

   Justification: All datasets, dimensions d, ranks r, segment counts K, horizon T, seed counts, probe rate, regularisation, window length, and per-baseline hyperparameters are listed in §5 and Appendix E; per-cell tables in Appendix F report mean± SE for every dataset, (d, r) cell, and method.

5. **Open access to data and code**

   Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

   Answer: [Yes]

   Justification: The supplemental archive contains all experiment scripts (one per figure and per table), per-dataset environment code, cached JSON results, and a README mapping each figure and table to its generator script.

6. **Experimental setting/details**

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   Answer: [Yes]

Justification: §5 states the headline parameters per benchmark; Appendix E gives the full hyperparameter table (probe period, ridge λ, window W, confidence δ, probe cost c, baseline-specific knobs) used uniformly across all (d, r) cells without per-dataset retuning.

7. **Experiment statistical significance**

   Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

   Answer: [Yes]

   Justification: Every reported number is mean$\pm$ SE or mean$\pm$ SEM over at least 10 (typically 10–30) seeds; figures show ± 1 SE bands; the seed count and aggregation choice are stated in each table caption and in §5.

8. **Experiments compute resources**

   Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

   Answer: [Yes]

   Justification: Appendix E reports that the full grid (all benchmarks, 10 seeds, 10 methods) takes 6–20 hours on a single 16-core CPU; no GPU is required (NumPy/SciPy only).

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   Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics `https://neurips.cc/public/EthicsGuidelines`?

   Answer: [Yes]

   Justification: The work is a theoretical and empirical study of a stochastic bandit algorithm on synthetic and publicly available benchmarks; the clinical environments are semi-synthetic surrogates calibrated to published cohort statistics, with no patient-identifiable data, no human subjects, and no clinical deployment.

10. **Broader impacts**

    Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

    Answer: [N/A]

    Justification: The paper studies a foundational sequential-decision algorithm with no specific deployment; the positive impact is improved sample efficiency for low-rank decision problems (recommendation, dosing) and negative-impact pathways are not direct. Clinical bandit benchmarks are evaluation surrogates, not deployable dosing policies.

11. **Safeguards**

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Justification: All datasets used (UCI Covertype, Pendigits, Satimage; MNIST; Fashion-MNIST; MovieLens-100K; Open Bandit Dataset [Saito et al., 2021]) and all baseline algorithms (LinUCB, D-LinUCB, SW-LinUCB, LowOFUL, VOFUL, LowRank-Reward, LinTS, BOSS, Jedra) are cited at first use; clinical benchmarks follow the IWPC [Consortium, 2009] and Rybak et al. [Rybak et al., 2020] dosing guidelines.

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    Answer: [Yes]

    Justification: The supplemental material releases the SPSC and SPSC-Adaptive Python implementations, the per-benchmark environments, all experiment scripts, and a README documenting the figure-to-script and table-to-source mappings.

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